



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD2523 - DATABASE

SEMESTER 1 2024/2025

PHASE 1– PROJECT PROPOSAL & DATABASE REQUIREMENT

GROUP 3 (PROJECT NAME: FUYOO)

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SECTION: 10

LECTURER: DR. LAYLA RASHEED ABDALLAH HASAN

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1.0 Introduction

Although nowadays there are various systems that are beneficial for improving business efficiency, many business owners still choose to conduct business using traditional methods, especially in the populations of older individuals or those who are not proficient in various technologies and systems. They operate their businesses traditionally. This approach leads to a lack of marketing and promotion for their businesses, often facing resource constraints with limited opportunities to access capital and resources. This mode of operation is no longer suitable for the modern business environment, as the development of the internet and digital technologies provides companies with new business opportunities and competitive advantages.

Therefore, we have decided to design a user-friendly system. This system will automate inventory management, providing real-time updates on stock levels. Furthermore, the system will assist in calculating profits by tracking sales and expenses, enabling the owners to monitor their financial performance effectively. By creating such a system, the business owners can streamline their operations and enhance overall efficiency in managing its fruit inventory and sales.

2.0 Background Study

The ABC fruit store used to sell tropical fruits, but Mr Lim, the seller, now wants to expand his business and start selling imported fruits like apples and strawberries. Since there is not a wide variety of tropical fruits sold before, Mr Lim carried out his business by using traditional methods like using a top-down management structure, fixed working hours, face-to-face sales interactions, and other traditional methods. He and his workers used manual methods to track inventory, such as counting fruits upon arrival and keeping records on paper.

With the introduction of imported fruits, Mr Lim's business is getting better as most of his customers are more likely to purchase the imported fruits. However, at the same time he realizes that the management is nearly out of control because of the inconvenience and low efficiency of using traditional methods.

Thus, he feels the need for a more organized system as he now has to categorize fruits, set prices accordingly, track sales and expenses, monitor his business financial performance and deal with other management in a proper way. To address this, Mr Lim is looking to implement a system that can help them to categorize fruits, differentiate between tropical and imported ones, set prices accordingly, and arrange them by price from cheapest to most expensive and other managements.

3.0 Problem Statement

Making the transition from selling all tropical fruits to imported fruits like apples and strawberries in its inventory, Mr. Lim's Fruit Store is facing significant challenges. The traditional methods of doing things include an established work schedule, a top-down management structure, and manual inventory tracking which cannot keep up with the demands of the growing company. Therefore, a number of urgent problems have emerged:

1. Ineffective Management Practices:

- Since imported fruits have been introduced, the use of traditional management techniques, such as manual inventory tracking and personal sales interactions, has grown more laborious and ineffective. The store's capacity to properly classify fruits, establish pricing, track sales and expenses, and maintain focus on overall business success is being affected by this inefficiency.

2. Ineffective Inventory Management:

- Traditional techniques, such as counting fruit as it arrives and keeping paper records, are capable of mistakes. Due to the lack of structure in the inventory management system, it may be challenging to distinguish between tropical fruits and imported fruits when the selection of fruits increases to include both. This could cause confusion and possibly lead to inventory shortages.

3. An Inadequate Method for Pricing:

- Mr. Lim discovered that it is challenging to establish prices for a variety of fruits in a way that fairly represents the fruit's value and consumer demand in the absence of a systematic pricing approach. It is challenging to rank fruits from least expensive to most expensive in the absence of a uniform pricing structure, which can result in lost sales opportunities and income disparities.

4. Limited Financial Visibility and Profit Calculations:

- The store's financial performance was not clear due to the manual recording of sales and expenses. Mr Lim did not have a system in place to compute sales revenue, total profit, or the cost of fruit that he bought. This made it more difficult for him to decide on price policies, inventory control, and general company planning. It was essential for Mr. Lim to put in place a method to compute sales and profits so he could see his financial performance clearly and adjust his strategic business decisions accordingly.

4.0 Proposed Solutions

Fruit sellers, such as Mr. Lim can find a reliable method of managing their inventory while monitoring sales and costs on just one platform with Fuyoo, a system designed specially to meet their needs. It has an easy-to-use interface and is also customizable to suit both small and large-scale fruit selling businesses.

The main purpose of the Fuyoo system is to introduce efficient inventory management for fruit sellers to record and track their fruit stock. The process of entering data is now automatically done, because fruit sellers can now use user-friendly forms to enter data into the system's database, replacing the manual counting and using paper to keep track of stocks.

Users can also sort and group fruits into different categories by using the data organizing features in Fuyoo. It makes it more convenient for fruit sellers to find the information of each type of fruit they sell including their source and prices. With Fuyoo, fruit sellers can locate specific items easier and manage their stock more efficiently.

Fuyoo also has functions for managing sales and tracking expenses. Fruit sellers can understand their financial situation better and allocate resources and budgets by keeping track of costs associated with buying fruits, maintaining their stores, and other operating expenses.

Fuyoo generates reports for fruit sellers which lets them learn important details about their business status. To maximize efficiency and profits, fruit sellers can plan their operations and make well-informed decisions with reports on trending fruit sales, profitability, and expenses. With the help of this system, fruit sellers like Mr. Lim can analyze sales and drive business growth and success by using all the tools offered.

4.1 Technical Feasibility

ABC Fruit Store has been using paper records to manually count and manage their stocks. This method is inefficient and may affect their business success. To implement the Fuyoo system, hardwares such as mobile devices and computers are required. Besides, a cloud based architecture is needed for accessibility from multiple locations over the internet. The store currently lacks the hardwares and technical support to manage their business. The costs for both software and hardware upgrades are within the budget. Thus, this system is technically feasible.

4.2 Operational Feasibility

The majority of employees of ABC Fruit Store are younger generations which are more familiar and proficient with technology. Thus, they possess the ability to adapt to a new system with minimal training. Targeted training sessions can be conducted for employees who are less experienced in this area to address possible issues they would encounter with managing the system. The business is currently well-positioned to transition to the new system in terms of operations.

4.3 Economical Feasibility

Cost Benefit Analysis

Estimated Costs	
Hardware	RM25,000
Software	RM20,000
Salary	RM25,000
Maintenance	RM5,000
Training	RM5,000

Expected Benefits	
Increase Sales	RM30,000
Savings	RM15,000

Assumptions	
Discount Rate	10%
Sensitivity Factor (Cost)	0.7
Sensitivity Factor (Benefit)	1.1
Annual Increment (Cost)	5%
Annual Increment (Benefit)	7%

Costs	Year 0	Year 1	Year 2	Year 3
-------	--------	--------	--------	--------

Development Costs				
- Hardware	17500			
- Software	14000			
Total	31500			
Production Costs				
- Salary		17500	18375	19294
- Maintenance		3500	3675	3859
- Training		5500	5775	6064
Annual Production Costs		26500	27825	29216
(Present Value)		24091	22996	21951
Accumulated Costs		55591	78587	78587

Benefits	Year 0	Year 1	Year 2	Year 3
- Increase Sales		33000	35310	37782
- Savings		16500	17655	18891
Annual Benefits		49500	52965	56673
(Present Value)		45000	43773	42579
Accumulated Benefits		45000	88773	131352
Gain or Loss		-10591	10186	52765
Profitability Index	1.68			

Profitability index (PI) is 1.68 which is larger than 1. This shows that it is a good investment and that it is feasible economically.

5.0 Objectives

- To identify and implement effective management practices
- To optimize inventory management
- To implement effective pricing strategies
- To enhance financial visibility and profit calculations

6.0 Scope

A) System

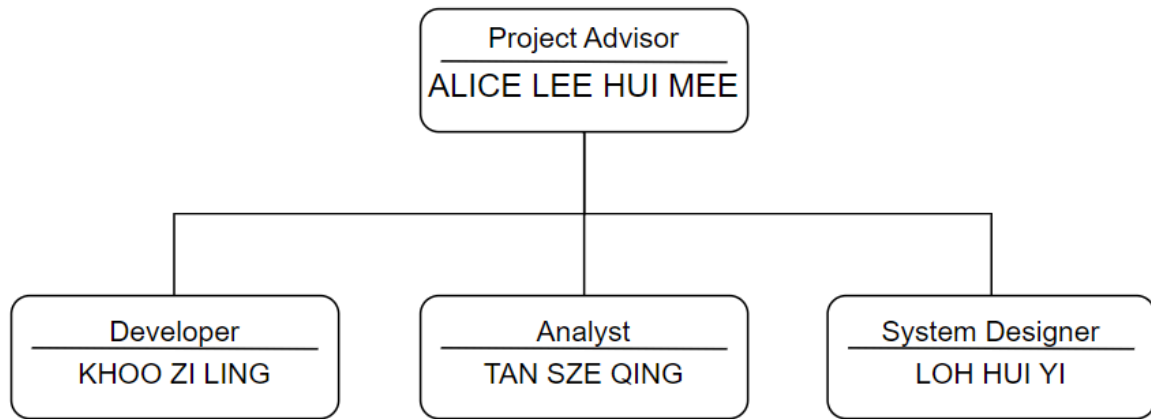
- Set up an inventory management system that can classify fruits(identify between imported and tropical kinds), and monitor stock levels in real time.
- Enable barcode scanning, automated data entry, and supplier database linkage.
- Provide pricing recommendations, create sales invoices, and monitor sales activities.
- Establish a financial management system to figure out the amount of fruit costs, how much money is made from sales, and how much profit is made all together.
- Provide financial reports and automate processes for calculating profits.

B) User- Sellers (Mr Lim and his workers)

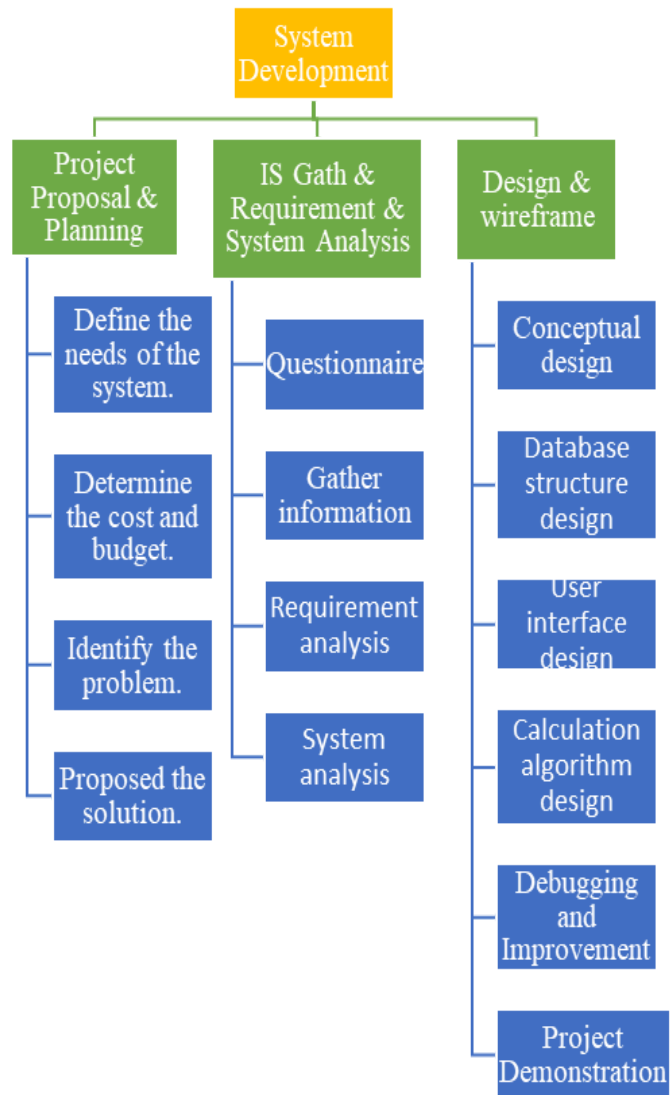
- View Fruit Information(each fruit in stock, including name, origin, variety, price, and quantity available).
- Update Fruit Details
- Adjust Inventory Level
- Price Management(to set and modify prices for different fruits based on factors such as market demand, quality, and seasonality, ensuring competitive pricing and maximizing revenue).
- Record and track sales, including the fruits sold, quantities, prices, and payment methods used.
- Monitor sales revenue, expenses, profit margins, and other key financial metrics, providing insights into the store's financial performance.
- Notification system of low stock levels, expired fruits, or other inventory-related issues.
- Data Backup and Recovery system for user
- User Authentication and Access Control

7.0 Project Planning

7.1 Human Resource

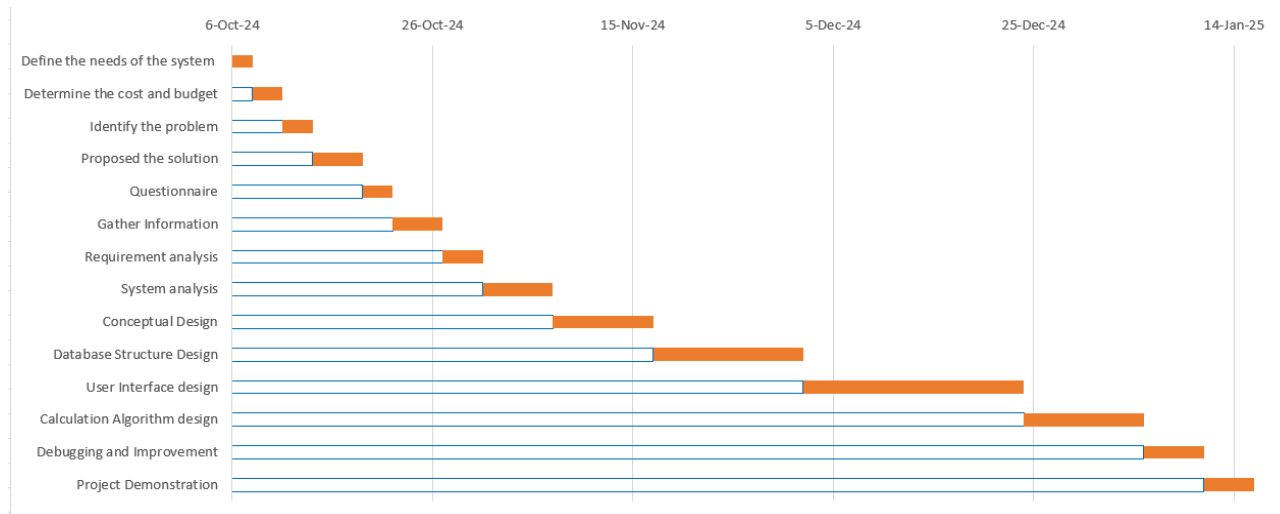


7.2 Work Breakdown Structure (WBS)



7.3 Gantt Chart

A	B	C	D	E	F
<u>Task</u>	<u>Activity</u>	<u>Predecessor</u>	<u>Duration (DAY)</u>	<u>Start Date</u>	<u>End Date</u>
A	Define the needs of the system	None	3	6-Oct-24	8-Oct-24
B	Determine the cost and budget	A	4	8-Oct-24	11-Oct-24
C	Identify the problem	B	4	11-Oct-24	14-Oct-24
D	Proposed the solution	C	6	14-Oct-24	19-Oct-24
E	Questionnaire	D	4	19-Oct-24	22-Oct-24
F	Gather Information	E	6	22-Oct-24	27-Oct-24
G	Requirement analysis	F	5	27-Oct-24	31-Oct-24
H	System analysis	G	8	31-Oct-24	7-Nov-24
I	Conceptual Design	H	11	7-Nov-24	17-Nov-24
J	Database Structure Design	I	16	17-Nov-24	2-Dec-24
K	User Interface design	J	23	2-Dec-24	24-Dec-24
L	Calculation Algorithm design	K	13	24-Dec-24	5-Jan-25
M	Debugging and Improvement	L	7	5-Jan-25	11-Jan-25
N	Project Demonstration	M	6	11-Jan-25	16-Jan-25



8.0 Requirement Analysis (based on AS-IS system form Scopes)

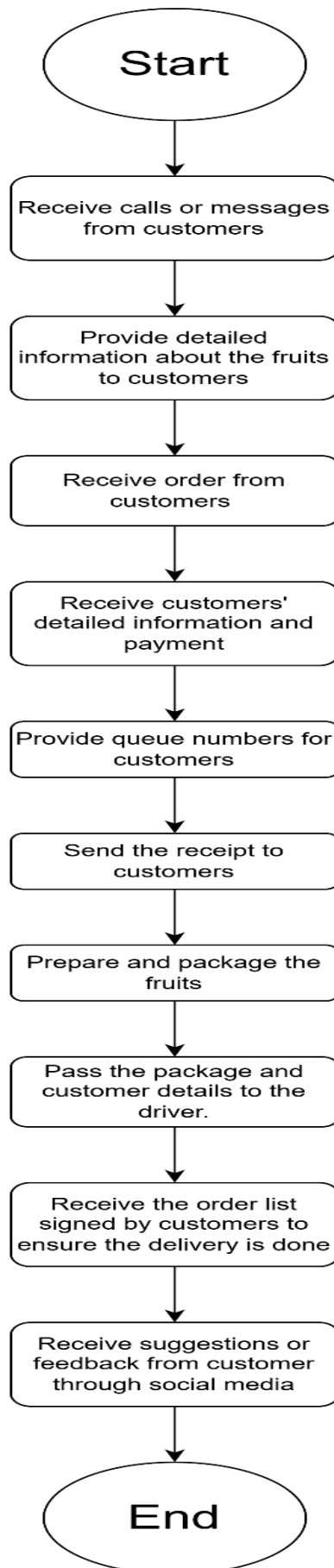
8.1 Current business process (scenarios, workflow)

ABC Fruit Store is a fruit wholesale company in Johor. This fruit store sells a wide variety of imported fruit and local fruit. The fruit store's system is currently using its Facebook page and Xiaohongshu page to promote their fruits. Customers can contact "ABC Fruit Store" through Whatsapp, phone call or Messenger to place the order and get more detailed information here. The entities that are involved in this system are the customer, seller and driver.

Firstly, customers will contact the fruit store using Whatsapp, phone call or Messenger. Then the seller will give the detailed information about the fruits such as the available stock, fruits price and when they will receive the stock. Customers will decide which type of fruit to place order and the quantity for each type. When customers place the order, they are required to make payment and give their detailed information such as phone number, name and delivery address. All this detailed information and payment made are required to send through whatsapp to the seller in order to notify them. After that, the seller will give a queue number for them and the receipt of the order.

When the stock of fruits is ready and packaging is done , the seller will give the package and customer details to the driver. Driver will send the fruits to customers' delivery addresses. When the stock is received by customers, they are required to sign on the order list given by the seller to the driver in order to confirm items received. Afterwards, the driver will give back the order list signed by customers to the seller to ensure delivery is done.

Customers can give suggestions or feedback through Whatsapp or Messenger so that ABC Fruit Store will have improvement for their system and products. If customers wish to place another order they just need to give their phone number and all the details given by previous such as name, delivery address, previous order already stored in the system.



9.0 Transaction requirement (data entry, data update/delete, data queries)

9.1 Data Entry

- Enter details about the customer like customer's name, contact number, email and so on.
- Enter detailed information about the fruits such as fruit types, fruit stock and fruit price.
- Enter the order by selecting fruit type and quantity.
- Enter the payment confirmation after receiving payment details like phone number, name and delivery address.
- Enter a queue number about the customer in the system.
- Enter the receipt about the customer.
- Enters details about the package and assigns it to a driver in the system once the fruit order is packaged.
- Enter the feedback or suggestions about the customer.

9.2 Data Update/Delete

- Update/delete order information from the system's database.
- Update/delete the customer's payment confirmation and queue number.
- Update/delete the customer's name, phone number, and delivery address.
- Update/delete order and package details.
- Update/delete fruit types, stock and price.
- Update/delete the payment status and receipt.
- Update/delete the order by selecting fruit type and quantity.
- Update/delete feedback and suggestions.

9.3 Data Queries

- Display the list of customer's details including name, contact number, and address.
- Display information about available fruit types, quantities, and prices.
- Display a list of all unprocessed orders.
- Display specific orders using the assigned queue number.
- Display sales reports for specific time periods.
- Display records of quality checks performed on the fruit before packaging to ensure quality and freshness.
- Display a list of pending deliveries assigned to each driver.
- Display current status of specific orders, including pending, packaged, delivered.
- Display customer feedback and suggestions for review.

10.0 Benefits and Summary of the Proposed System

The proposed system will ensure ease of use for the ABC fruit store's staff. The system will automate inventory management processes which will reduce manual tasks and avoid human error. Real-time updates on stock levels ensure accurate and up-to-date data on the fruits' inventory, enabling better decision-making and inventory control. Besides, the system will help to calculate the earnings by tracking the sales and expenses in order to show important information about the financial health of ABC fruit stores.

Furthermore, this system streamlines operations by centralizing inventory and sales data which can minimize an abundance of errors such as human errors. This is because it can provide accurate and precise data. Meanwhile, this user-friendly system provides a simplicity of use of the system enhancing the overall user experience.

All in all, the implementation of this system will undoubtedly save time and enhance the operational efficiency of ABC fruit stores.

Staff's benefits:

- Improved inventory management efficiency
- Avoid human error
- Reduce wasting time
- Increase the convenience

Manager's benefits:

- Save time by completing the financial report analysis
- Digital records reduce the amount of unnecessary paperwork
- Easy access to the profit and loss
- More accurate decisions can be made based on the system

11.0 Summary

By completing the database requirements document of ABC fruit shop, our team has a deeper understanding of the current system operation. They allow consumers to place fruit orders and ask questions via a variety of methods, including Whatsapp, phone calls, and Messenger.

Customers receive all the information they require regarding what is on hand and when it will be delivered. After making a decision, they use WhatsApp to share the location of delivery and pay. Subsequently, the vendor collaborates with the drivers to guarantee that the products reach the clients on schedule. Customers sign a list provided by the driver to verify they received the fruit when it comes. Additionally, customers can use Whatsapp or Messenger to provide suggestions and feedback to help the business continue to improve.

In a nutshell, the demand of this collection process makes us have a broader understanding of the current database system of fruit enterprises. Comprehensive documentation and in-depth analysis provide us with many concepts of data and programs. Therefore, we will use the knowledge collected from this method to help us create a more efficient platform.



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PHASE2 -DATABASE CONCEPTUAL DESIGN (ERD)

NAME	MATRIC NUMBER
ALICE LEE HUI MEE	A23CS0040
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5.0 Data Dictionary

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6.0 Summary

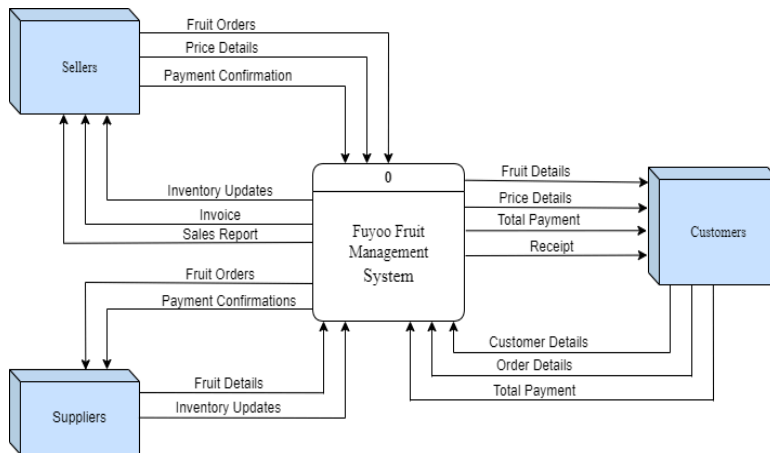
1.0 Introduction

Although nowadays there are various systems that are beneficial for improving business efficiency, many business owners still choose to conduct business using traditional methods, especially in the populations of older individuals or those who are not proficient in various technologies and systems. They operate their businesses traditionally. This approach leads to a lack of marketing and promotion for their businesses, often facing resource constraints with limited opportunities to access capital and resources. This mode of operation is no longer suitable for the modern business environment, as the development of the internet and digital technologies provides companies with new business opportunities and competitive advantages.

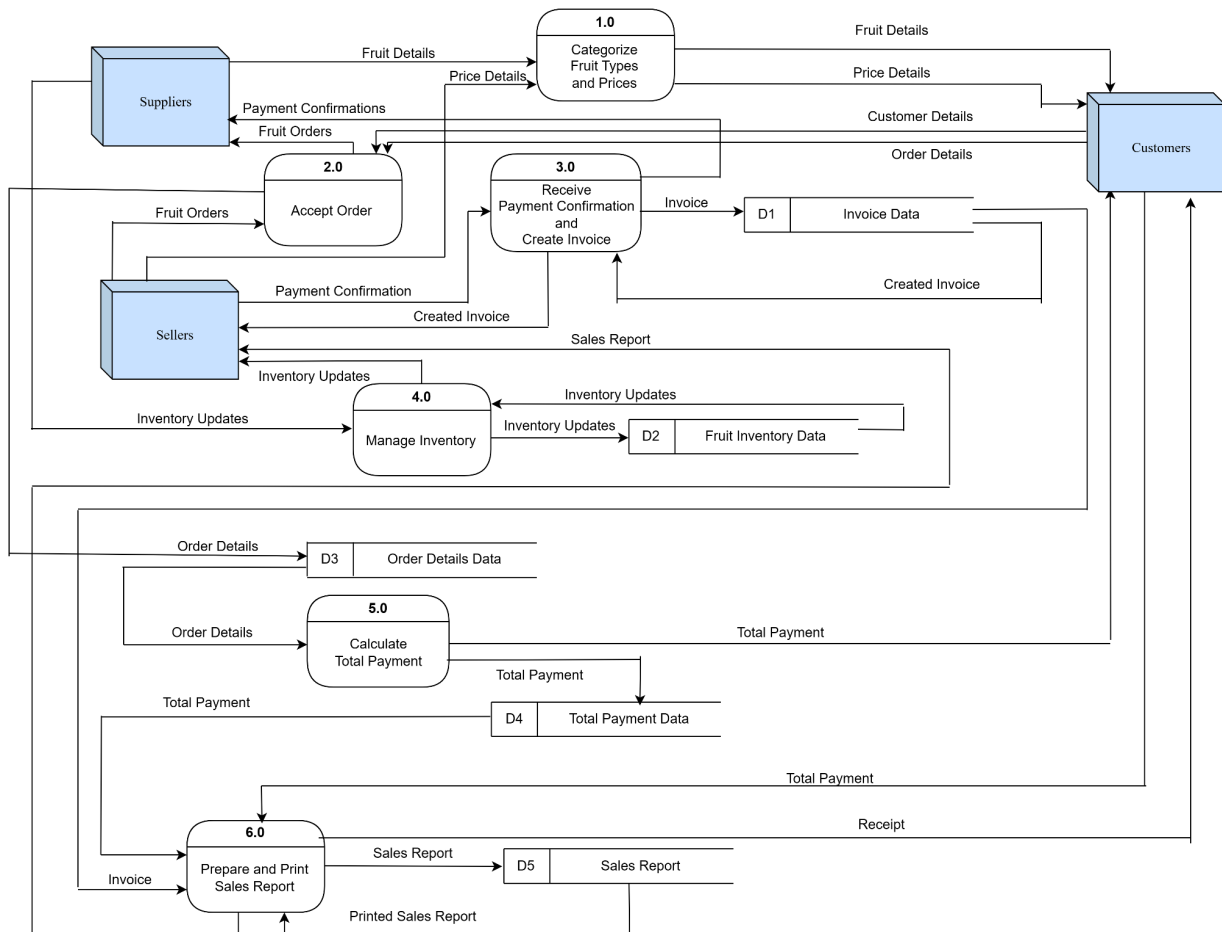
Therefore, we have decided to design a user-friendly system. This system will automate inventory management, providing real-time updates on stock levels. Furthermore, the system will assist in calculating profits by tracking sales and expenses, enabling the owners to monitor their financial performance effectively. By creating such a system, the business owners can streamline their operations and enhance overall efficiency in managing its fruit inventory and sales.

2.0 Data Flow Diagram (DFD)

2.1 Context Diagram

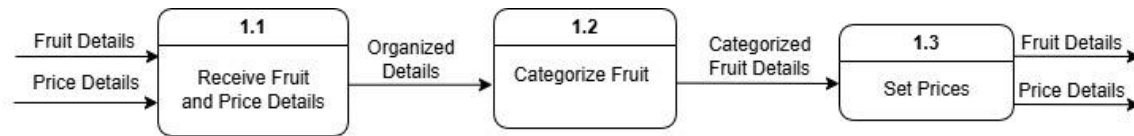


2.2 Parent Diagram (Level 0)

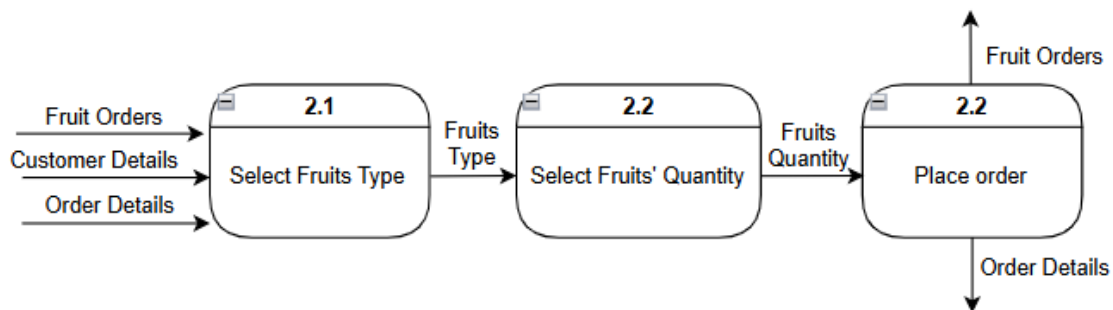


2.3 Child Diagram

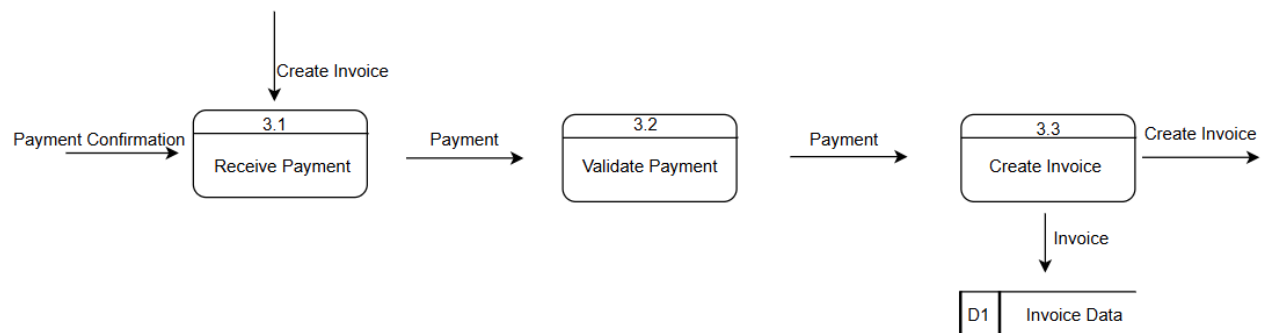
3.3.1 Process 1.0 <Categorize Fruit Types and Prices>



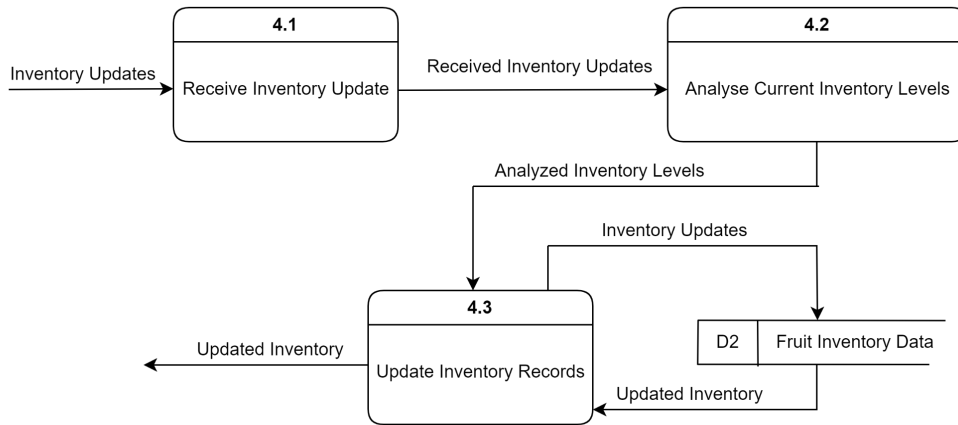
3.3.2 Process 2.0 <Accept Order>



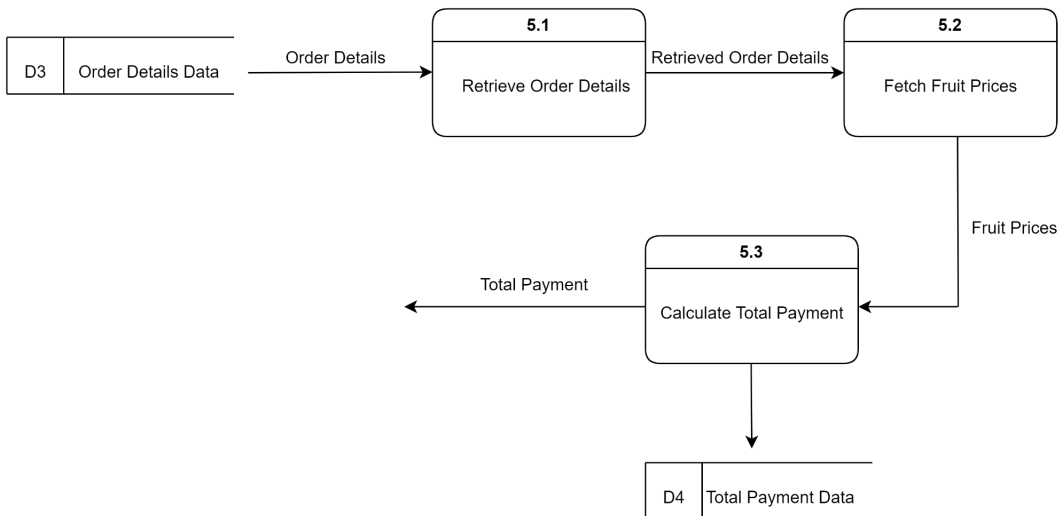
3.3.3 Process 3.0 <Receive Order Confirmation and Create Invoice>



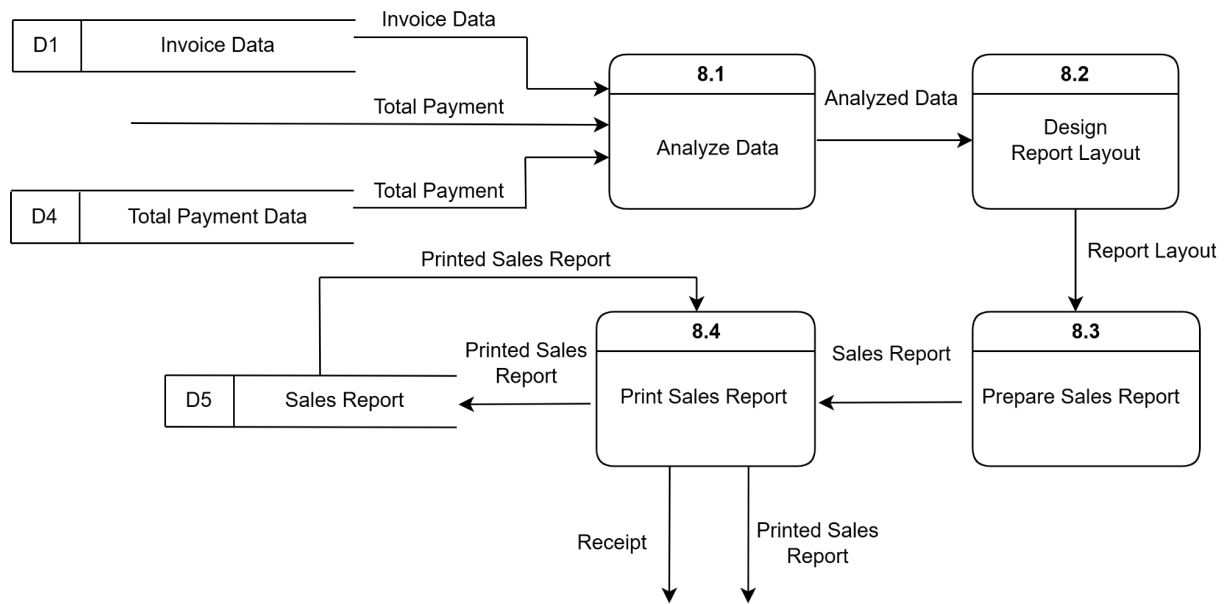
3.3.4 Process 4.0 <Manage Inventory>



3.3.5 Process 5.0 <Calculate Total Payment>



3.3.6 Process 6.0 <Prepare and Print Sales Report>



3.0 Data & Transaction Requirement

3.1 Proposed business rule

Customer:

- View fruit's type, quantity, and price in the system.
- Place an order by selecting the type and quantity of fruit in the system.
- Make online payment for order confirmation in the system.
- Receive a digital receipt from the seller after the online payment is successful.
- Track order status digitally through notifications in the system.
- Provide feedback or suggestions through the system.

Seller:

- Enter and update new details about fruit types, stock and prices in the system.
- Record customer details, orders, payments, and assigned queue numbers in the system.
- Process orders digitally by selecting the fruit types and quantities.
- Generates and sends digital order receipts to customers and records in the system.
- Assign fruit's package to a driver digitally for delivery.
- Store and review customer's feedback or suggestions digitally in the system.

Suppliers:

- Provide sellers with regular digital updates on fruit's quantity, prices, and delivery times.
- Notify sellers of changes in fruit stock or prices through the system.
- Generate a digital invoice for each shipment sent to the seller.
- Receive digital notifications from sellers about upcoming orders.

Driver:

- Receive packed fruits and customer's detail from the seller digitally.
- Deliver the fruits to the customer and update the delivery status in the system.
- Obtain an e-signature from the customer upon delivery to confirm receipt.
- Update the seller digitally on any issues encountered during the delivery.

3.2 Proposed data & transactional

3.2.1 Proposed Data Requirement

Customer

The data stored in the Customer entity includes customer ID, customer name, contact number, and address. The customer ID is defined as the primary key. The name and the address are the composite attributes. The name consists of first name and last name, whereas the address consists of street, city and postcode. A customer can place zero or more orders. A customer can provide zero or more feedback after making their purchase. A customer can make one or more payments. A customer is only in charge by one seller.

Seller

The data stored in the Seller entity includes seller ID, seller name, contact number, email, location, customer ID, supplier ID and invoice ID. The seller ID is defined as the primary key as it is unique. Customer ID, supplier ID and invoice ID will be used as the foreign key, with reference to the Customer entity, Supplier entity and Invoice entity. The name consists of first name and last name. A seller can receive zero or more payments. A seller can receive zero or more invoices. A seller will manage zero or more inventories. A seller can make one or more payments.

Supplier

The data stored in the Supplier entity includes supplier ID, supplier name, contact number, email, and address. The supplier ID is defined as the primary key because of its uniqueness. The name and the address are the composite attributes. The name consists of first name and last name while the address consists of street, city and postcode. The supplier can receive one or more than one payment. A supplier can send zero or more than one invoice. A supplier can manage zero or many sellers.

Driver

The data stored in the Driver entity consists of the driver ID, driver name, contact details and vehicle details. The driver ID is defined as the primary key. A driver can deliver zero or more than one order.

Order

The data stored in the Order entity consists of the order ID, fruit type, order date, total amount, payment status, queue number, customer ID and driver ID. The primary key for the Order entity is the order ID. There are two foreign keys which are customer ID and driver ID by referencing

the Customer entity and Driver entity. An order is placed by only one customer. An order is delivered by only one driver.

Payment

The data kept in the Payment entity are payment ID, amount to be paid, payment date, payment method, customer ID and seller ID. The payment ID is the primary key for the Payment entity. Both customer ID and seller ID are the foreign keys with reference from Customer entity and Seller entity. A payment is received by only one supplier. A payment is received by only one seller. A payment is made by only one seller for their purchase. A payment is made by only one supplier.

Inventory

The data saved in the Inventory entity consists of inventory ID, fruit type, the available quantity of fruits and seller ID. Because of its uniqueness, the inventory ID is defined as the primary key. The inventory is managed by only one seller.

Invoice

The data kept in the Invoice entity are invoice ID, invoice date, total amount, order ID and seller ID. The invoice ID is selected as the primary key. The order ID and seller ID are defined as the foreign key reference from the Order entity and Seller entity. An invoice is received by only one seller. An invoice is sent by only one supplier to the seller.

Feedback

The data stored in the Feedback entity includes feedback ID, message, date and customer ID. The primary key for the entity of Feedback is feedback ID. As it needs to trace the satisfaction purchase experience from customers, it needs the customer ID as a foreign key to get the feedback from each customer. Feedback is provided by only one customers based on their purchasing experience.

3.2.2 Proposed Transactional Requirement

Data Entry

- Enter details about the customer like customer's name, contact number, email and so on.
- Enter detailed information about the fruits such as fruit types, fruit stock and fruit price.
- Enter the order by selecting fruit type and quantity.
- Enter the payment confirmation after receiving payment details like phone number, name and delivery address.

- Enter a queue number about the customer in the system.
- Enter the receipt about the customer.
- Enters details about the package and assigns it to a driver in the system once the fruit order is packaged.
- Enter the feedback or suggestions about the customer.

Data Update/Delete

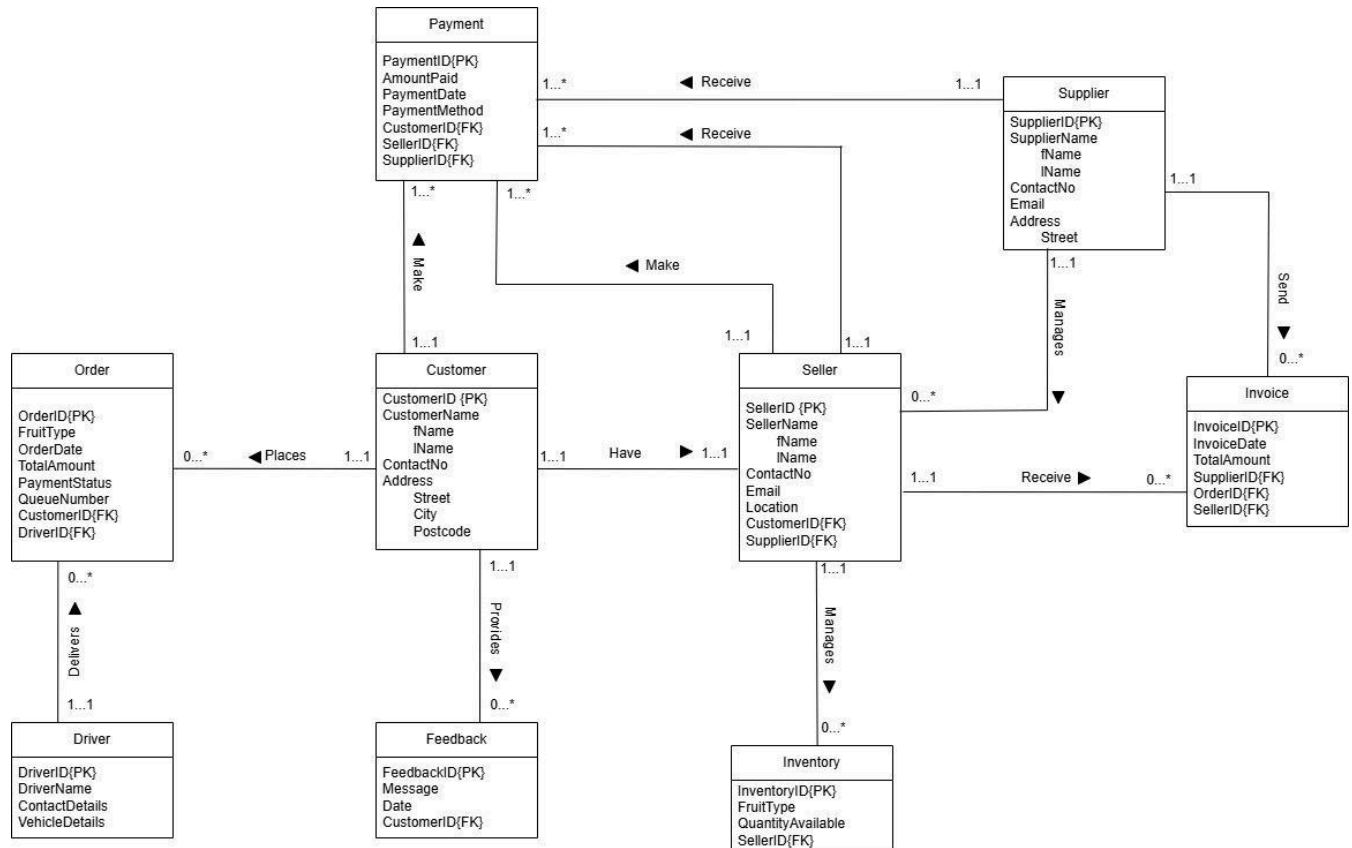
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- Update/delete the customer's payment confirmation and queue number.
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- Update/delete order and package details.
- Update/delete fruit types, stock and price.
- Update/delete the payment status and receipt.
- Update/delete the order by selecting fruit type and quantity.
- Update/delete feedback and suggestions.

Data Queries

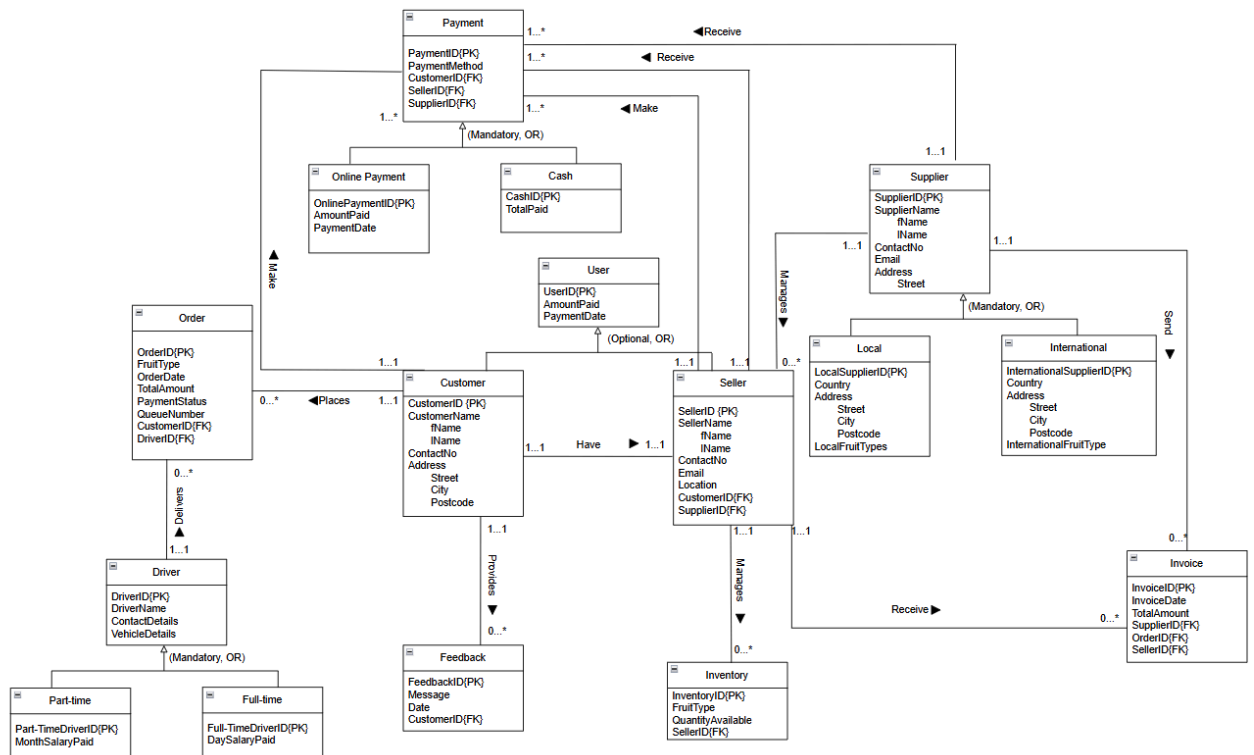
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- Display records of quality checks performed on the fruit before packaging to ensure quality and freshness.
- Display a list of pending deliveries assigned to each driver.
- Display current status of specific orders, including pending, packaged, delivered.
- Display customer feedback and suggestions for review.

4.0 Database Conceptual Design

4.1 Conceptual ERD



4.2 Enhanced ERD (EERD)



5.0 Data Dictionary

5.1 Description of Entity

Entity	Description	Occurrence
Customer	Customer's detail	The customer placed orders, made the payment and gave feedback.
Seller	Seller's detail	The seller receives the order placed by the customer and maintains records of available fruits, orders, and payment
Supplier	Supplier's detail	Provide the fruits to the seller
Order	Order's detail	Represent the transactions between customers and sellers. Each order includes details like fruit type, amount, and pricing
Payment	Payment's detail	Payment is generated when the customer pays for their orders and the seller pays for their inventory orders placed
Inventory	Inventory's detail	The total current stock of fruits is managed by the seller and each item in inventory is identified by fruit type and quantity available
Invoice	Invoice's detail	Invoices are generated for seller orders, detailing the amount due and order information
Driver	Driver's detail	Deliver the fruits ordered by customer.
Feedback	Feedback's detail	Customer returns their opinions based on their experience buying the fruits.

5.2 Description of Relationship

Entity	Multiplicity	Relationship	Multiplicity	Entity
Customer	1...*	Places	0...*	Order
Customer	1...1	Provides	0...*	Feedback
Customer	1...1	Have	1...1	Seller
Customer	1...*	Make	0...*	Payment
Seller	1...1	Receives	0...*	Payment
Seller	1...1	Receive	0...*	Invoice
Seller	1...1	Manages	0...*	Inventory
Seller	1...1	Makes	1...*	Payment
Supplier	1...1	Receive	1...*	Payment
Supplier	1...1	Send	0...*	Invoice
Supplier	1...1	Manages	0...*	Seller
Driver	1...1	Delivers	0...*	Order

5.3 Description Attributes

Entity	Attributes	Description	Data Type	Null	Multi-valued
Customer	CustomerID	Identity of customer (PK)	VARCHAR2(10)	No	No
	CustomerName	Name of customer	VARCHAR2(40)	No	No
	fNAME	First name of customer	VARCHAR2(20)	No	No
	lName	Last name of customer	VARCHAR2(20)	No	No
	ContactNo	Contact number of customer	VARCHAR2(12)	No	Yes
	Address	Address of customer	VARCHAR2(50)	No	No

	Street	Street address of customer	VARCHAR2(20)	No	No
	City	City address of customer	VARCHAR2(15)	No	No
	Postcode	Postcode address of customer	VARCHAR2(5)	No	No
Seller	SellerID	Identity of seller (PK)	VARCHAR2(10)	No	No
	SellerName	Name of seller	VARCHAR2(40)	No	No
	fName	First name of seller	VARCHAR2(20)	No	No
	lName	Last name of seller	VARCHAR2(20)	No	No
	ContactNo	Contact number of seller	VARCHAR2(12)	No	Yes
	Email	Email of seller	VARCHAR	No	Yes
	Location	Location of seller	VARCHAR2(50)	No	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR2(10)	No	No
	SupplierID	Foreign key for identity of supplier (FK)	VARCHAR2(10)	No	No
	InvoiceID	Foreign key for identity of invoice (FK)	VARCHAR2(10)	No	No
Supplier	SupplierID	Identity of supplier (PK)	VARCHAR2(10)	No	No
	SupplierName	Name of supplier	VARCHAR2(40)	No	No
	fName	First name of supplier	VARCHAR2(	No	No

			10)		
	lName	Last name of supplier	VARCHAR2(10)	No	No
	ContactNo	Contact number of supplier	VARCHAR2(12)	No	Yes
	Email	Email of supplier	VARCHAR	No	Yes
	Address	Address of supplier	VARCHAR2(50)	No	No
	Street	Street address of supplier	VARCHAR2(20)	No	No
	City	City address of supplier	VARCHAR2(15)	No	No
	Postcode	Postcode address of supplier	VARCHAR2(5)	No	No
Driver	DriverID	Identity of driver (PK)	VARCHAR2(10)	No	No
	DriverName	Name of driver	VARCHAR2(40)	No	No
	ContactDetails	Contact details of driver	VARCHAR	No	Yes
	VehicleDetails	Details of driver's vehicle	VARCHAR	No	No
Order	OrderID	Identity of order (PK)	VARCHAR2(10)	No	No
	FruitType	Type of fruit	VARCHAR2(15)	Yes	Yes
	OrderDate	Date of order placed by customer	TIMESTAMP	No	Yes
	TotalAmount	Total amount need to be paid by customer	FLOAT(10)	No	No
	PaymentStatus	Payment status of order	VARCHAR2(10)	Yes	No

	QueueNumber	Queue number of customer	VARCHAR2(10)	Yes	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR2(10)	No	No
	DriverID	Foreign key for identity of driver (FK)	VARCHAR2(10)	No	No
Payment	PaymentID	Identity of payment (PK)	VARCHAR2(10)	No	No
	AmountPaid	Amount paid by customer	FLOAT(10)	Yes	No
	PaymentDate	Payment date of order	TIMESTAMP	Yes	No
	PaymentMethod	Payment method	VARCHAR2(15)	No	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR2(10)	No	No
	SellerID	Foreign key for identity of seller (FK)	VARCHAR2(10)	No	No
Inventory	InventoryID	Identity of inventory (PK)	VARCHAR2(10)	No	No
	FruitType	Type of fruit	VARCHAR2(15)	Yes	Yes
	QuantityAvailable	Quantity of fruits that are available in stock	VARCHAR2(5)	No	Yes
	SellerID	Foreign key for identity of seller (FK)	VARCHAR2(10)	No	No
Invoice	InvoiceID	Identity of invoice (PK)	VARCHAR2(10)	No	No
	InvoiceDate	Date of invoice	TIMESTAMP	No	No
	TotalAmount	Total amount paid by customer	FLOAT(10)	No	No
	OrderID	Foreign key for	VARCHAR2(	No	No

		identity of order (FK)	10)		
	SellerID	Foreign key for identity of seller (FK)	VARCHAR2(10)	No	No
Feedback	FeedbackID	Identity of feedback (PK)	VARCHAR2(10)	No	No
	Message	Message of feedback given by customer	VARCHAR2	Yes	Yes
	Date	Date of feedback sent	TIMESTAMP	Yes	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR2(10)	No	No

6.0 Summary

In phase 2, we successfully created a database structure design for the ABC fruit store system. Through this process, we gained valuable insights into the operational needs of the fruit wholesale business and how the structured database system can be improved. Initially, we update the proposed business rules so that they meet the system requirements, and identify data and transaction processes for smooth operations. Also, a data flow diagram (DFD) with a context diagram, a parent diagram (diagram 0) and a child diagram was created to recognize how information works in the system, including processes such as ordering, updating inventory, confirming payments, making delivery and so on.

Additionally, conceptual entity relationship diagrams (ERD) and enhanced ERD are created to show the relationship and multiplicity between entities, namely customers, sellers, suppliers, and drivers. These two diagrams serve to improve the structure of the data and ensure every important relationship is included. Next, the data dictionary provides clear description and data types for each entity and attribute, which help us ensure consistency and accuracy in our designs.

To conclude, this phase acts as an important step in building a database for the ABC fruit store system. This is because it provides a solid foundation to improve efficiency and provide better service for customers. Within this design in place, the system can meet current needs as well as handle future growth. It makes the ABC fruit store more organized and provides a better experience for everyone.



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD2523 - DATABASE

SEMESTER 1 2024/2025

PHASE 3–DATABASE LOGICAL DESIGN & SQL

GROUP 3 (PROJECT NAME: FUYOO)

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1.0 Introduction

Although nowadays there are various systems that are beneficial for improving business efficiency, many business owners still choose to conduct business using traditional methods, especially in the populations of older individuals or those who are not proficient in various technologies and systems. They operate their businesses traditionally. This approach leads to a lack of marketing and promotion for their businesses, often facing resource constraints with limited opportunities to access capital and resources. This mode of operation is no longer suitable for the modern business environment, as the development of the internet and digital technologies provides companies with new business opportunities and competitive advantages.

Therefore, we have decided to design a user-friendly system. This system will automate inventory management, providing real-time updates on stock levels. Furthermore, the system will assist in calculating profits by tracking sales and expenses, enabling the owners to monitor their financial performance effectively. By creating such a system, the business owners can streamline their operations and enhance overall efficiency in managing its fruit inventory and sales.

2.0 Overview of project

In the phase of database logical design, we will convert the conceptual Entity Relationship Diagrams produced in Phase 2 into logical ERD. For the fruit business management system, modifications will be made to address the complexities of inventory and sales management. This will prepare the foundation for building a structured and efficient database. Each entity in the logical ERD will be converted into a relational schema, with attributes defined and primary keys created for each table.

For the fruit business system, entities such as customer, seller, supplier, payment, order and so on need to be clearly defined and interrelated. Normalization of our system will also take place during this stage by eliminating redundancies and dependencies in the derived relational schema. For example, ensuring that inventory details are not duplicated across records and that sales transactions are correctly associated with specific fruit items and financial records will help maintain data integrity. At the same time, the data dictionary will be updated as necessary to reflect changes brought about by normalization.

Lastly, we will create SQL statements to verify logical ERD with the interface design and transaction needs of the system. This is necessary for ensuring that the database structure facilitates fundamental functions including financial reporting, profit computations, sales tracking, and inventory changes.

3.0 Database conceptual design

3.1 Updated business rule

Customer:

- View fruit's type, quantity, and price in the system.
- Place an order by selecting the type and quantity of fruit in the system.
- Make online payment for order confirmation in the system.
- Receive a digital receipt from the seller after the online payment is successful.
- Track order status digitally through notifications in the system.
- Provide feedback or suggestions through the system.

Seller:

- Enter and update new details about fruit types, stock and prices in the system.
- Record customer details, orders, payments, and assigned queue numbers in the system.
- Process orders digitally by selecting the fruit types and quantities.
- Generates and sends digital order receipts to customers and records in the system.
- Assign fruit's package to a driver digitally for delivery.
- Store and review customer's feedback or suggestions digitally in the system.

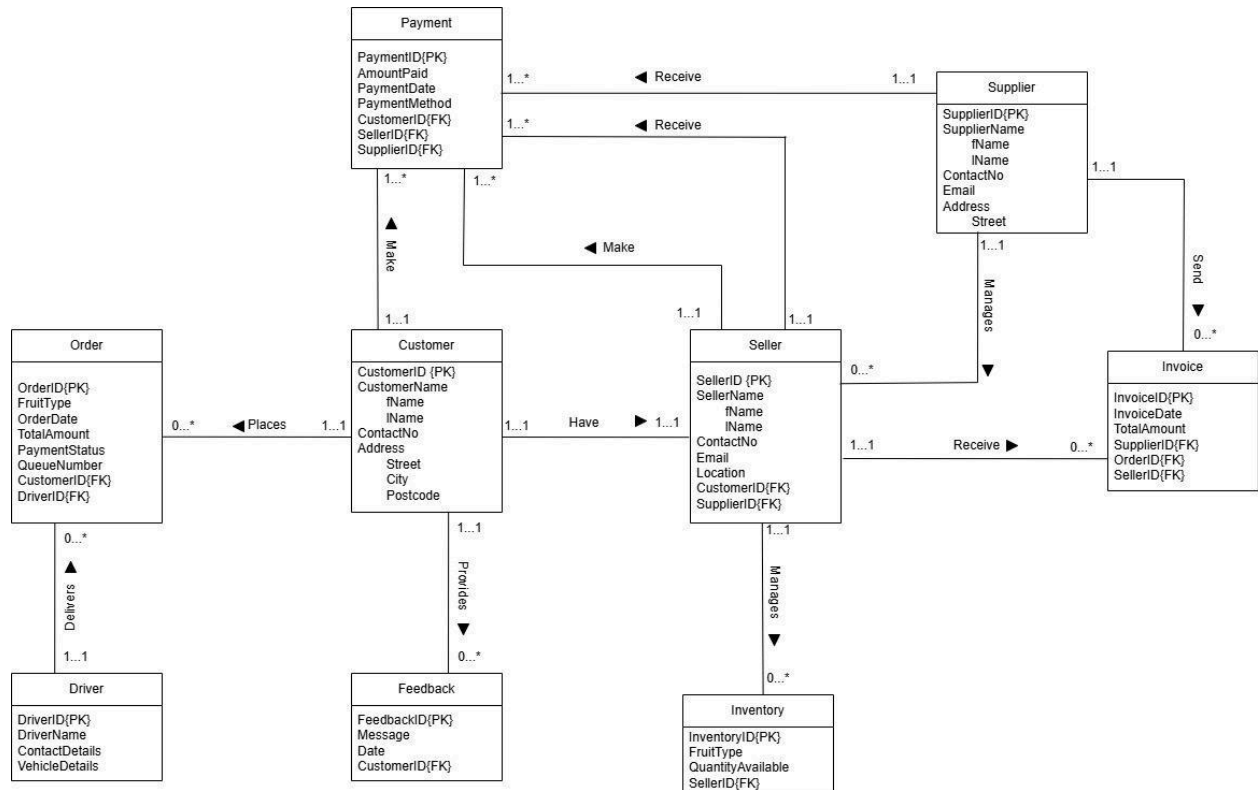
Suppliers:

- Provide sellers with regular digital updates on fruit's quantity, prices, and delivery times.
- Notify sellers of changes in fruit stock or prices through the system.
- Generate a digital invoice for each shipment sent to the seller.
- Receive digital notifications from sellers about upcoming orders.

Driver:

- Receive packed fruits and customer's detail from the seller digitally.
- Deliver the fruits to the customer and update the delivery status in the system.
- Obtain an e-signature from the customer upon delivery to confirm receipt.
- Update the seller digitally on any issues encountered during the delivery.

3.2 Conceptual ERD



4.2 Updated Data Dictionary

Supplier	Supplier's detail	Provide the fruits to the seller .
Local	Detail of supplier who sells local fruits	Provide local fruits to the seller.
International	Detail of supplier who sells imported fruits	Provide imported fruit to the seller.
Order	Order's detail	Represent the transactions between customers and sellers. Each order includes details like fruit type, amount, and pricing.
Payment	Payment's detail	Payment is generated when the customer pays for their orders and the seller pays for their inventory orders placed.
Online Payment	Payment's detail when paying online	One of the option for customer and seller to make payment
Cash	Payment's detail when paying by cash	One of the option for customer and seller to make payment
Inventory	Inventory's detail	The total current stock of fruits is managed by the seller and each item in inventory is identified by fruit type and quantity available.
Invoice	Invoice's detail	Invoices are generated for seller orders, detailing the amount due and order information.
Driver	Driver's detail	Deliver the fruits ordered by the customer.
Part-time	Part-time driver's detail	Records of salary of part-time driver
Full-time	Full-time driver's detail	Records of salary of full-time driver

Feedback	Feedback's detail	Customer returns their opinions based on their experience buying the fruits.
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Description of Relationship

Entity	Multiplicity	Relationship	Multiplicity	Entity
Customer	1...*	Places	0...*	Order
Customer	1...1	Provides	0...*	Feedback
Customer	1...1	Have	1...1	Seller
Customer	1...*	Make	0...*	Payment
Customer	0...1	Part of	1...1	User
Seller	1...1	Receives	0...*	Payment
Seller	1...1	Receive	0...*	Invoice
Seller	1...1	Manages	0...*	Inventory
Seller	1...1	Makes	1...*	Payment
Seller	0...1	Part of	1...1	User
Supplier	1...1	Receive	1...*	Payment
Supplier	1...1	Send	0...*	Invoice
Supplier	1...1	Manages	0...*	Seller
Local	0...1	Part of	1...1	Supplier
International	0...1	Part of	1...1	Supplier
Driver	1...1	Delivers	0...*	Order
Part-time	0...1	Part of	1...1	Driver
Full-time	0...1	Part of	1...1	Driver
Online Payment	0...1	Part of	1...1	Payment
Cash	0...1	Part of	1...1	Payment

Description of Attributes

Entity	Attributes	Description	Data Type	Null	Multi-valued
User	UserID	Identity of user (PK)	VARCHAR(10)	No	No
	AmountPaid	Amount paid by user	FLOAT(10)	No	No
	PaymentDate	Date of amount paid by user	TIMESTAMP	No	Yes
Customer	CustomerID	Identity of customer (PK)	VARCHAR(10)	No	No
	CustomerName	Name of customer	VARCHAR(40)	No	No
	fNAME	First name of customer	VARCHAR(20)	No	No
	lName	Last name of customer	VARCHAR(20)	No	No
	ContactNo	Contact number of customer	VARCHAR(12)	No	Yes
	Address	Address of customer	VARCHAR(50)	No	No
	Street	Street address of customer	VARCHAR(20)	No	No
	City	City address of customer	VARCHAR(15)	No	No
	Postcode	Postcode address of customer	VARCHAR(5)	No	No
Seller	SellerID	Identity of seller (PK)	VARCHAR(10)	No	No
	SellerName	Name of seller	VARCHAR(40)	No	No
	fName	First name of seller	VARCHAR(20)	No	No
	lName	Last name of seller	VARCHAR(20)	No	No

			0)		
	ContactNo	Contact number of seller	VARCHAR(12)	No	Yes
	Email	Email of seller	VARCHAR	No	Yes
	Location	Location of seller	VARCHAR(50)	No	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR(10)	No	No
	SupplierID	Foreign key for identity of supplier (FK)	VARCHAR(10)	No	No
	InvoiceID	Foreign key for identity of invoice (FK)	VARCHAR(10)	No	No
Supplier	SupplierID	Identity of supplier (PK)	VARCHAR(10)	No	No
	SupplierName	Name of supplier	VARCHAR(40)	No	No
	fName	First name of supplier	VARCHAR(10)	No	No
	lName	Last name of supplier	VARCHAR(10)	No	No
	ContactNo	Contact number of supplier	VARCHAR(12)	No	Yes
	Email	Email of supplier	VARCHAR	No	Yes
	Address	Address of supplier	VARCHAR(50)	No	No
	Street	Street address of supplier	VARCHAR(20)	No	No
Local	LocalSupplierID	Identity of supplier that provides local fruits to seller (PK)	VARCHAR(10)	No	No

	Country	Name of origin country of the fruit	VARCHAR(20)	No	No
	Address	Address of supplier that provides local fruits to seller	VARCHAR(50)	No	No
	Street	Street address of supplier that provides local fruits to seller	VARCHAR(20)	No	No
	City	City address of supplier that provides local fruits to seller	VARCHAR(15)	No	No
	Postcode	Postcode address of supplier that provides local fruits to seller	VARCHAR(5)	No	No
	LocalFruitTypes	Types of local fruits	VARCHAR(15)	Yes	Yes
International	InternationalSupplierID	Identity of supplier that provides imported fruits to seller (PK)	VARCHAR (10)	No	No
	Country	Name of origin country of the fruit	VARCHAR(20)	No	No
	Address	Address of supplier that provides imported fruits to seller	VARCHAR(50)	No	No
	Street	Street address of supplier that provides imported fruits to seller	VARCHAR(20)	No	No
	City	City address of supplier that provides imported fruits to seller	VARCHAR(15)	No	No
	Postcode	Postcode address of supplier that provides imported fruits to seller	VARCHAR(5)	No	No

	InternationalFruitType	Types of imported fruits	VARCHAR2(15)	Yes	Yes
Driver	DriverID	Identity of driver (PK)	VARCHAR(10)	No	No
	DriverName	Name of driver	VARCHAR(40)	No	No
	ContactDetails	Contact details of driver	INT(12)	No	Yes
	VehicleDetails	Details of driver's vehicle	VARCHAR	No	No
Part-time	Part-TimeDriverID	Identity of part-time driver (PK)	VARCHAR(10)	No	No
	MonthSalaryPaid	Monthly salary of part-time driver	FLOAT(10)	No	No
Full-time	Full-TimeDriverID	Identity of full-time driver (PK)	VARCHAR(10)	No	No
	DaySalaryPaid	Day salary of full-time driver	FLOAT(10)	No	No
Order	OrderID	Identity of order (PK)	VARCHAR(10)	No	No
	FruitType	Type of fruit	VARCHAR(15)	Yes	Yes
	OrderDate	Date of order placed by customer	TIMESTAMP	No	Yes
	TotalAmount	Total amount need to be paid by customer	FLOAT(10)	No	No
	PaymentStatus	Payment status of order	VARCHAR(10)	Yes	No
	QueueNumber	Queue number of customer	VARCHAR(10)	Yes	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR(10)	No	No

	DriverID	Foreign key for identity of driver (FK)	VARCHAR(10)	No	No
Payment	PaymentID	Identity of payment (PK)	VARCHAR(10)	No	No
	PaymentMethod	Payment method	VARCHAR(15)	No	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR(10)	No	No
	SellerID	Foreign key for identity of seller (FK)	VARCHAR(10)	No	No
	SupplierID	Foreign key for identity of supplier (FK)	VARCHAR(10)	No	No
Online Payment	OnlinePaymentID	Identity of online payment (PK)	VARCHAR(10)	No	No
	AmountPaid	Amount paid by customer through online transaction	FLOAT(10)	Yes	No
	PaymentDate	Payment date of order	TIMESTAMP	Yes	No
Cash	CashID	Identity of cash payment (PK)	VARCHAR(10)	No	No
	TotalPaid	Amount paid by customer through cash	FLOAT(10)	Yes	No
Inventory	InventoryID	Identity of inventory (PK)	VARCHAR(10)	No	No
	FruitType	Type of fruit	VARCHAR(15)	Yes	Yes
	QuantityAvailable	Quantity of fruits that are available in stock	VARCHAR(5)	No	Yes
	SellerID	Foreign key for identity of seller (FK)	VARCHAR(10)	No	No
Invoice	InvoiceID	Identity of invoice	VARCHAR(10)	No	No

		(PK)	0)		
	InvoiceDate	Date of invoice	TIMESTAMP	No	No
	TotalAmount	Total amount paid by customer	FLOAT(10)	No	No
	OrderID	Foreign key for identity of order (FK)	VARCHAR(10)	No	No
	SellerID	Foreign key for identity of seller (FK)	VARCHAR(10)	No	No
Feedback	FeedbackID	Identity of feedback (PK)	VARCHAR(10)	No	No
	Message	Message of feedback given by customer	VARCHAR	Yes	Yes
	Date	Date of feedback sent	TIMESTAMP	Yes	No
	CustomerID	Foreign key for identity of customer (FK)	VARCHAR(10)	No	No

4.3 Normalization

1. USER (UserID, AmountPaid, PaymentDate)

fd1: UserID $\rightarrow$ UserID, AmountPaid, PaymentDate

1NF&2NF&3NF&BCNF:

USER (UserID, AmountPaid, PaymentDate)

2. CUSTOMER (CustomerID, CustomerName, fName, lName, ContactNo, Address, Street, City, Postcode)

fd1: CustomerID $\rightarrow$ CustomerName, fName, lName, ContactNo, Address, Street, City, Postcode

1NF&2NF&3NF&BCNF:

CUSTOMER (CustomerID, CustomerName, fName, lName, ContactNo, Address, Street, City, Postcode)

3. SELLER (SellerID, SellerName, fName, lName, ContactNo, Email, Location, CustomerID, SupplierID, InvoiceID)

fd1: SellerID $\rightarrow$ SellerName, fName, lName, ContactNo, Email, Location, CustomerID, SupplierID, InvoiceID

1NF&2NF&3NF&BCNF:

SELLER (SellerID, SellerName, fName, lName, ContactNo, Email, Location, CustomerID, SupplierID, InvoiceID)

4. SUPPLIER (SupplierID, SupplierName, fName, lName, ContactNo, Email, Address, Street)

fd1: SupplierID $\rightarrow$ SupplierName, fName, lName, ContactNo, Email, Address, Street

1NF&2NF&3NF&BCNF:

SUPPLIER (SupplierID, SupplierName, fName, lName, ContactNo, Email, Address, Street)

5. LOCAL (LocalSupplierID, Country, Address, Street, City, Postcode, LocalFruitTypes)

fd1: LocalSupplierID → Country, Address, Street, City, Postcode, LocalFruitTypes

1NF&2NF&3NF&BCNF:

LOCAL (LocalSupplierID, Country, Address, Street, City, Postcode, LocalFruitTypes)

6. INTERNATIONAL (InternationalSupplierID, Country, Address, Street, City, Postcode, InternationalFruitType)

fd1: InternationalSupplierID → Country, Address, Street, City, Postcode, InternationalFruitType

1NF&2NF&3NF&BCNF:

INTERNATIONAL (InternationalSupplierID, Country, Address, Street, City, Postcode, InternationalFruitType)

7. DRIVER (DriverID, DriverName, ContactDetails, VehicleDetails)

fd1: DriverID → DriverName, ContactDetails, VehicleDetails

1NF&2NF&3NF&BCNF:

DRIVER (DriverID, DriverName, ContactDetails, VehicleDetails)

8. PART-TIME (Part-TimeDriverID, DaySalaryPaid)

fd1: Part-TimeDriverID → DaySalaryPaid

1NF&2NF&3NF&BCNF:

PART-TIME (Part-TimeDriverID, DaySalaryPaid)

9. FULL-TIME (Full-TimeDriverID, MonthSalaryPaid)

fd1: Full-TimeDriverID → MonthSalaryPaid

1NF&2NF&3NF&BCNF:

FULL-TIME (Full-TimeDriverID, MonthSalaryPaid)

10. ORDER (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)

fd1: DriverID → DriverName, ContactDetails, VehicleDetails

1NF&2NF&3NF&BCNF:

ORDER (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)

11. PAYMENT (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)

fd1: PaymentID → PaymentMethod, CustomerID, SellerID, SupplierID

1NF&2NF&3NF&BCNF:

PAYMENT (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)

12. ONLINE PAYMENT (OnlinePaymentID, AmountPaid, PaymentDate)

fd1: OnlinePaymentID → AmountPaid, PaymentDate

1NF&2NF&3NF&BCNF:

ONLINE PAYMENT (OnlinePaymentID, AmountPaid, PaymentDate)

13. CASH (CashID, TotalPaid)

fd1: CashID → TotalPaid

1NF&2NF&3NF&BCNF:

CASH (CashID, TotalPaid)

14. INVENTORY (InventoryID, FruitType, QuantityAvailable, SellerID)

fd1: InventoryID $\rightarrow$ FruitType, QuantityAvailable, SellerID

1NF&2NF&3NF&BCNF:

INVENTORY (InventoryID, FruitType, QuantityAvailable, SellerID)

15. INVOICE (InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)

fd1: InvoiceID $\rightarrow$ InvoiceDate, TotalAmount, OrderID, SellerID

1NF&2NF&3NF&BCNF:

INVOICE (InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)

16. FEEDBACK (FeedbackID, Message, Date, CustomerID)

fd1: FeedbackID $\rightarrow$ Message, Date, CustomerID

1NF&2NF&3NF&BCNF:

FEEDBACK (FeedbackID, Message, Date, CustomerID)

Remark: The underlined words are primary keys.

5.0 Relational DB Schemas (after normalization)

This section illustrates a set of relation database schema for the Fuyoo system.

User	(<u>UserID</u> , AmountPaid, PaymentDate)
Customer	(<u>CustomerID</u> , CustomerName, fName, lName, ContactNo, Address, Street, City, Postcode)
Seller	(<u>SellerID</u> , SellerName, fName, lName, ContactNo, Email, Location, CustomerID, SupplierID, InvoiceID)
Supplier	(<u>SupplierID</u> , SupplierName, fName, lName, ContactNo, Email, Address, Street)
Local	(<u>LocalSupplierID</u> , Country, Address, Street, City, Postcode, LocalFruitTypes)
International	(<u>InternationalSupplierID</u> , Country, Address, Street, City, Postcode, InternationalFruitType)
Driver	(<u>DriverID</u> , DriverName, ContactDetails, VehicleDetails)
Part-time	(<u>Part-TimeDriverID</u> , DaySalaryPaid)
Full-time	(<u>Full-TimeDriverID</u> , MonthSalaryPaid)
Order	(<u>OrderID</u> , FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)
Payment	(<u>PaymentID</u> , PaymentMethod, CustomerID, SellerID, SupplierID)
Online payment	(<u>OnlinePaymentID</u> , AmountPaid, PaymentDate)
Cash	(<u>CashID</u> , TotalPaid)
Inventory	(<u>InventoryID</u> , FruitType, QuantityAvailable, SellerID)
Invoice	(<u>InvoiceID</u> , InvoiceDate, TotalAmount, OrderID, SellerID)
Feedback	(FeedbackID, Message, Date, CustomerID)

Remark: The underlined words are primary keys.

6.0 SQL Statements (DDL & DML)

6.1 DDL

/*Create table for each entity*/

```
CREATE TABLE User(  
    UserID VARCHAR(10) PRIMARY KEY,  
    AmountPaid FLOAT(10,2) NOT NULL,  
    PaymentDate DATETIME NOT NULL  
);
```

```
CREATE TABLE Customer(  
    CustomerID VARCHAR(10) PRIMARY KEY,  
    CustomerName VARCHAR(40) NOT NULL,  
    fNAME VARCHAR(20) NOT NULL,  
    lName VARCHAR(20) NOT NULL,  
    ContactNo VARCHAR(12) NOT NULL,  
    Address VARCHAR(50) NOT NULL,  
    Street VARCHAR(20) NOT NULL,  
    City VARCHAR(15) NOT NULL,  
    Postcode VARCHAR2(5) NOT NULL  
);
```

```
CREATE TABLE Seller(  
    SellerID VARCHAR(10) PRIMARY KEY,  
    SellerName VARCHAR(40) NOT NULL,  
    fNAME VARCHAR(20) NOT NULL,  
    lName VARCHAR(20) NOT NULL,  
    ContactNo VARCHAR(12) NOT NULL,  
    Address VARCHAR(50) NOT NULL,  
    Street VARCHAR(20) NOT NULL,  
    City VARCHAR(15) NOT NULL,  
    Postcode VARCHAR2(5) NOT NULL  
);
```

SellerID VARCHAR(10) PRIMARY KEY,
SellerName VARCHAR(40) NOT NULL,
fName VARCHAR(20) NOT NULL,
lName VARCHAR(20) NOT NULL,
ContactNo VARCHAR(12) NOT NULL,
Email VARCHAR(50) NOT NULL,
Location VARCHAR(50) NOT NULL,
CustomerID VARCHAR(10) NOT NULL,
SupplierID VARCHAR(10) NOT NULL,
InvoiceID VARCHAR(10) NOT NULL,
FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),
FOREIGN KEY (InvoiceID) REFERENCES Invoice(InvoiceID)
);

CREATE TABLE Supplier(
SupplierID VARCHAR(10) PRIMARY KEY,
SupplierName VARCHAR(40) NOT NULL,
fName VARCHAR(10) NOT NULL,
lName VARCHAR(10) NOT NULL,
ContactNo VARCHAR(12) NOT NULL,
Email VARCHAR(50) NOT NULL,
Address VARCHAR2(50) NOT NULL,

```
Street VARCHAR(20) NOT NULL,  
City VARCHAR(15) NOT NULL,  
Postcode VARCHAR(5) NOT NULL  
);
```

```
CREATE TABLE Local(  
    LocalSupplierID VARCHAR(10) PRIMARY KEY,  
    Country VARCHAR(20) NOT NULL,  
    Street VARCHAR(20) NOT NULL,  
    City VARCHAR(15) NOT NULL,  
    Postcode VARCHAR(5) NOT NULL,  
    LocalFruitTypes VARCHAR(15)  
);
```

```
CREATE TABLE International(  
    InternationalSupplierID VARCHAR(10) PRIMARY KEY,  
    Country VARCHAR(20) NOT NULL,  
    Street VARCHAR(20) NOT NULL,  
    City VARCHAR(15) NOT NULL,  
    Postcode VARCHAR(5) NOT NULL,  
    InternationalFruitType VARCHAR(15)  
);
```

```
CREATE TABLE Driver(  
    DriverID VARCHAR(10) PRIMARY KEY,  
    DriverName VARCHAR(40) NOT NULL,  
    ContactDetails VARCHAR(12) NOT NULL,  
    VehicleDetails VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE PartTime (  
    PartTimeDriverID VARCHAR(10) PRIMARY KEY,  
    DaySalaryPaid FLOAT(10) NOT NULL  
);
```

```
CREATE TABLE FullTime (  
    FullTimeDriverID VARCHAR(10) PRIMARY KEY,  
    MonthSalaryPaid FLOAT(10) NOT NULL  
);
```

```
CREATE TABLE Order (  
    OrderID VARCHAR(10) PRIMARY KEY,  
    FruitType VARCHAR(15),  
    OrderDate TIMESTAMP NOT NULL,  
    TotalAmount FLOAT(10) NOT NULL,  
    PaymentStatus VARCHAR(10),
```

```
QueueNumber VARCHAR(10),  
CustomerID VARCHAR(10) NOT NULL,  
DriverID VARCHAR(10) NOT NULL,  
FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),  
FOREIGN KEY (DriverID) REFERENCES Driver(DriverID)  
);
```

```
CREATE TABLE Payment (  
    PaymentID VARCHAR(10) PRIMARY KEY,  
    AmountPaid FLOAT(10) NOT NULL,  
    PaymentDate TIMESTAMP NOT NULL,  
    PaymentMethod VARCHAR(15) NOT NULL,  
    CustomerID VARCHAR(10) NOT NULL,  
    SellerID VARCHAR2(10) NOT NULL,  
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),  
    FOREIGN KEY (SellerID) REFERENCES Seller(SellerID)  
);
```

```
CREATE TABLE Online Payment (  
    OnlinePaymentID VARCHAR(10) PRIMARY KEY,  
    AmountPaid FLOAT(10),  
    PaymentDate TIMESTAMP  
);
```

```
CREATE TABLE Cash(  
    CashID VARCHAR(10) PRIMARY KEY,  
    TotalPaid FLOAT(10)  
);
```

```
CREATE TABLE Inventory(  
    InventoryID VARCHAR(10) PRIMARY KEY,  
    FruitType VARCHAR(15),  
    QuantityAvailable VARCHAR(5) NOT NULL,  
    SellerID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (SellerID) REFERENCES Seller(SellerID)  
);
```

```
CREATE TABLE Invoice(  
    InvoiceID VARCHAR(10) PRIMARY KEY,  
    InvoiceDate TIMESTAMP NOT NULL,  
    TotalAmount FLOAT(10) NOT NULL,  
    OrderID VARCHAR(10) NOT NULL,  
    SellerID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (OrderID) REFERENCES Order(OrderID),  
    FOREIGN KEY (SellerID) REFERENCES Seller(SellerID)  
);
```

```
CREATE TABLE Feedback(  
    FeedbackID VARCHAR(10) PRIMARY KEY,  
    Message VARCHAR(225),  
    Date TIMESTAMP,  
    CustomerID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (CustomerID) REFERENCES Customer(SellerID)  
);
```

```
ALTER TABLE Supplier  
DROP COLUMN City,  
DROP COLUMN Postcode;
```

```
ALTER TABLE Payment  
DROP COLUMN AmountPaid,  
DROP COLUMN PaymentDate;
```

```
ALTER TABLE Payment  
ADD CONSTRAINT fk_supplier_ID FOREIGN KEY (SupplierID)  
REFERENCES Supplier(SupplierID);
```

6.2 DML

-User

```
INSERT INTO User(UserID, AmountPaid, PaymentDate)
```

```
VALUES
```

```
('U001', 150.50, TO_TIMESTAMP('2024-12-20 14:30:00', 'YYYY-MM-DD HH24:MI:SS')),
```

```
('U002', 230.00, TO_TIMESTAMP('2024-12-20 15:00:00', 'YYYY-MM-DD HH24:MI:SS')),
```

```
('U003', 120.75, TO_TIMESTAMP('2024-12-20 15:30:00', 'YYYY-MM-DD HH24:MI:SS'));
```

–View User table

```
SELECT * FROM User;
```

UserID	AmountPaid	PaymentDate
U001	150.50	2024-12-20 14:30:00
U002	230.00	2024-12-20 15:00:00
U003	120.75	2024-12-20 15:30:00

– Change AmountPaid

```
UPDATE User
```

```
SET AmountPaid = 180.75
```

```
WHERE UserID = 'U003';
```

–View the User table after update

```
SELECT * FROM User;
```

UserID	AmountPaid	PaymentDate
U001	150.50	2024-12-20 14:30:00
U002	230.00	2024-12-20 15:00:00
U003	180.75	2024-12-20 15:30:00

-Customer

INSERT INTO Customer (CustomerID, CustomerName, fName, lName, ContactNo, Street, City, Postcode, Address)

VALUES

('C001', 'John Tan', 'John', 'Tan', '0121231234', 'Jalan Lily', 'Johor Bahru', '81100', '123 Jalan Lily'),

('C002', 'Jane Lee', 'Jane', 'Lee', '0198887766', 'Jalan Rose', 'Kuala Lumpur', '50200', '456 Jalan Rose'),

('C003', 'Emily Chong', 'Emily', 'Chong', '01110088765', 'Jalan Hibiscus', 'Kuala Lumpur', '58200', '789 Jalan Hibiscus');

-View Customer table

SELECT * FROM Customer;

CustomerID	CustomerName	fName	lName	ContactNo	Address	Street	City	Postcode
C001	John Tan	John	Tan	0121231234	123 Jalan Lily	Jalan Lily	Johor Bahru	81100
C002	Jane Lee	Jane	Lee	0198887766	456 Jalan Rose	Jalan Rose	Kuala Lumpur	50200
C003	Emily Chong	Emily	Chong	01110088765	789 Jalan Hibiscus	Jalan Hibiscus	Kuala Lumpur	58200

- Change ContactNo

UPDATE Customer

SET ContactNo = '0123456999'

WHERE CustomerID = 'C002';

-View the Customer table after update

SELECT * FROM Customer;

CustomerID	CustomerName	fName	lName	ContactNo	Address	Street	City	Postcode
C001	John Tan	John	Tan	0121231234	123 Jalan Lily	Jalan Lily	Johor Bahru	81100
C002	Jane Lee	Jane	Lee	0123456999	456 Jalan Rose	Jalan Rose	Kuala Lumpur	50200
C003	Emily Chong	Emily	Chong	01110088765	789 Jalan Hibiscus	Jalan Hibiscus	Kuala Lumpur	58200

-Seller

INSERT INTO Seller(SellerID, SellerName, fName, lName, ContactNo, Email, Location, CustomerID, SupplierID, InvoiceID)

VALUES

('SE001', 'Happy Seller', 'Anna', 'Taylor', '0125556666', 'annataylor@hotmail.com', 'Downtown Market', 'C001', 'S001', 'I001'),

('SE002', 'Fresh Seller', 'Chris', 'Ngeow', '0174445555', 'chrisngeow@yahoo.com', 'City Square', 'C002', 'S002', 'I002'),

('SE003', 'Fruit Hub', 'Kate', 'Pang', '0143334444', 'katepang@gmail.com', 'Market Place', 'C003', 'S003', 'I003');

-View Seller table

SELECT * FROM Seller;

SellerID	SellerName	fName	lName	ContactNo	Email	Location	CustomerID	SupplierID	InvoiceID
SE001	Happy Seller	Anna	Taylor	0125556666	annataylor@hotmail.com	Downtown Market	C001	S001	I001
SE002	Fresh Seller	Chris	Ngeow	0174445555	chrisngeow@yahoo.com	City Square	C002	S002	I002
SE003	Fruit Hub	Kate	Pang	0143334444	katepang@gmail.com	Market Place	C003	S003	I003

- Change Email

UPDATE Seller

SET Email = 'ChrisNemail@gmail.com'

WHERE SellerID = 'SE002';

–View the Seller table after update

SELECT * FROM Seller;

SellerID	SellerName	fName	lName	ContactNo	Email	Location	CustomerID	SupplierID	InvoiceID
SE001	Happy Seller	Anna	Taylor	0125556666	annataylor@hotmail.com	Downtown Market	C001	S001	I001
SE002	Fresh Seller	Chris	Ngeow	0174445555	ChrisNemail@gmail.com	City Square	C002	S002	I002
SE003	Fruit Hub	Kate	Pang	0143334444	katepang@gmail.com	Market Place	C003	S003	I003

-Supplier

INSERT INTO Supplier (SupplierID, SupplierName, fName, lName, ContactNo, Email, Address, Street, City, Postcode)

VALUES

('S001', 'Fresh Fruits Co.', 'Mark', 'Lee', '0183344534', 'marklee@gmail.com', '123 Orchard Lane', 'Jalan Awan', 'Kuala Lumpur', '50000'),

('S002', 'Tropical Delights', 'Susan', 'Kim', '0167772222', 'susankim@hotmail.com', '456 Mango Road', 'Jalan Bintang', 'Johor Bahru', '81100'),

('S003', 'Berry World', 'James', 'Bond', '01387654321', 'jamesbond@gmail.com', '789 Berry Blvd', 'Jalan Bulan', 'Penang', '10200');

–View Supplier table

SELECT * FROM Supplier;

SupplierID	SupplierName	fName	lName	ContactNo	Email	Address	Street	City	Postcode
S001	Fresh Fruits Co.	Mark	Lee	0183344534	marklee@gmail.com	123 Orchard Lane	Jalan Awan	Kuala Lumpur	50000
S002	Tropical Delights	Susan	Kim	0167772222	susankim@hotmail.com	456 Mango Road	Jalan Bintang	Johor Bahru	81100
S003	Berry World	James	Bond	01387654321	jamesbond@gmail.com	789 Berry Blvd	Jalan Bulan	Penang	10200

-Local

INSERT INTO Local(LocalSupplierID, Country, Street, City, Postcode, LocalFruitTypes)

VALUES

('L001', 'Malaysia', 'Jalan Alor', 'Kuala Lumpur', '50200', 'Durian'),

('L002', 'Malaysia', 'Jalan Adda', 'Johor Bahru', '81100', 'Mangosteen'),

('L003', 'Malaysia', 'Jalan Resak', 'Johor Bahru', '81310', 'Papaya');

–View Local table

SELECT * FROM Local;

LocalSupplierID	Country	Street	City	Postcode	LocalFruitTypes
L001	Malaysia	Jalan Alor	Kuala Lumpur	50200	Durian
L002	Malaysia	Jalan Adda	Johor Bahru	81100	Mangosteen
L003	Malaysia	Jalan Resak	Johor Bahru	81310	Papaya

-International

INSERT INTO International (InternationalSupplierID, Country, Street, City, Postcode, InternationalFruitType)

VALUES

('I001', 'Kazakhstan', 'Snorlax St', 'Koksai', '040921', 'Apple'),

('I002', 'Sweden', 'Bygget 24', 'Surefors', '590 54', 'Strawberry'),

('I003', 'Philippines', 'Rizal Street', 'Poblacion', '1210 2', 'Starfruit');

–View International table

SELECT * FROM International;

InternationalSupplierID	Country	Street	City	Postcode	InternationalFruitType
I001	Kazakhstan	Snorlax St	Koksai	040921	Apple
I002	Sweden	Bygget 24	Surefors	590 54	Strawberry
I003	Philippines	Rizal Street	Poblacion	1210 2	Starfruit

–Driver

INSERT INTO Driver (DriverID, DriverName, ContactDetails, VehicleDetails)

VALUES ('D001', 'Andy Liu', '01234567890', 'Proton Vira');

```
INSERT INTO Driver (DriverID, DriverName, ContactDetails, VehicleDetails)
VALUES ('D002', 'Brayan Lee', '01234567891', 'Proton Saga');
```

```
INSERT INTO Driver (DriverID, DriverName, ContactDetails, VehicleDetails)
VALUES ('D003', 'Charmen Tee', '01234567890', 'Proton Persona');
```

–View Driver table

```
SELECT * FROM Driver;
```

```
mysql> SELECT * FROM Driver;
+-----+-----+-----+-----+
| DriverID | DriverName | ContactDetails | VehicleDetails |
+-----+-----+-----+-----+
| D001     | Andy Liu   | 01234567890    | Proton Vira    |
| D002     | Brayan Lee | 01234567891    | Proton Saga    |
| D003     | Charmen Tee | 01234567890    | Proton Persona |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

– Change contact number

```
UPDATE Driver
```

```
SET ContactDetails = 01188888888
```

```
WHERE DriverID = 'D003';
```

–View the Driver table after update

```
SELECT * FROM Driver;
```

```
mysql> SELECT * FROM Driver;
```

DriverID	DriverName	ContactDetails	VehicleDetails
D001	Andy Liu	01234567890	Proton Vira
D002	Brayan Lee	01234567891	Proton Saga
D003	Charmen Tee	1188888888	Proton Persona

```
3 rows in set (0.00 sec)
```

–Part-time

```
INSERT INTO PartTime (PartTimeDriverID, DaySalaryPaid)
```

```
VALUES ('PT001', 48);
```

```
INSERT INTO PartTime (PartTimeDriverID, DaySalaryPaid)
```

```
VALUES ('PT002', 64);
```

–View Part-time table

```
SELECT * FROM PartTime;
```

```
mysql> SELECT * FROM PartTime;
```

PartTimeDriverID	DaySalaryPaid
PT001	48
PT002	64

```
2 rows in set (0.00 sec)
```

–Full-time

```
INSERT INTO FullTime (FullTimeDriverID, MonthSalaryPaid)
```

```
VALUES ('FT001', 1600);
```

```
INSERT INTO FullTime (FullTimeDriverID, MonthSalaryPaid)
```

```
VALUES ('FT002', 1800);
```

```
INSERT INTO "FullTime" (FullTimeDriverID, MonthSalaryPaid)
```

```
VALUES ('FT003', 2000);
```

–View Full-time table

```
SELECT * FROM "FullTime";
```

```
mysql> SELECT * FROM FullTime;
+-----+-----+
| FullTimeDriverID | MonthSalaryPaid |
+-----+-----+
| FT001           | 1600           |
| FT002           | 1800           |
| FT003           | 2000           |
+-----+-----+
3 rows in set (0.00 sec)
```

–Place an order

```
INSERT INTO Order (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus,
QueueNumber, CustomerID, DriverID)
```

```
VALUES ('O001', 'apple', SYSDATE(), 12.00, 'successful', 'Q001', 'C001', 'D001');
```

```
INSERT INTO Order (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)
```

```
VALUES ('O002', 'melon', SYSDATE(), 20.00, 'successful', 'Q002', 'C002', 'D002');
```

```
INSERT INTO Order (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)
```

```
VALUES ('O003', 'berries', SYSDATE(), 22.00, 'successful', 'Q003', 'C001', 'D001');
```

```
INSERT INTO Order (OrderID, FruitType, OrderDate, TotalAmount, PaymentStatus, QueueNumber, CustomerID, DriverID)
```

```
VALUES ('O004', 'pears', SYSDATE(), 15.00, 'unsuccessful', 'Q004', 'C003', 'D003');
```

–View the order table

```
SELECT * FROM Order;
```

```
mysql> SELECT * FROM `Order`;
```

OrderID	FruitType	OrderDate	TotalAmount	PaymentStatus	QueueNumber	CustomerID	DriverID
O001	apple	2024-12-22 00:01:53	12.00	successful	Q001	C001	D001
O002	melon	2024-12-22 00:05:00	20.00	successful	Q002	C002	D002
O003	berries	2024-12-22 00:05:00	22.00	successful	Q003	C001	D001
O004	pears	2024-12-22 00:07:45	15.00	unsuccessful	Q004	C003	D003

4 rows in set (0.00 sec)

–Update an order being canceled

```
UPDATE Order
```

```
SET PaymentStatus = "Cancelled"
```

```
WHERE OrderID = 'O003';
```

–View the Order table after update

SELECT * FROM Order;

```
mysql>
mysql> SELECT * FROM `Order`;
+-----+-----+-----+-----+-----+-----+-----+-----+
| OrderID | FruitType | OrderDate | TotalAmount | PaymentStatus | QueueNumber | CustomerID | DriverID |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 0001 | apple | 2024-12-22 00:01:53 | 12.00 | successful | Q001 | C001 | D001 |
| 0002 | melon | 2024-12-22 00:05:00 | 20.00 | successful | Q002 | C002 | D002 |
| 0003 | berries | 2024-12-22 00:05:00 | 22.00 | Cancelled | Q003 | C001 | D001 |
| 0004 | pears | 2024-12-22 00:07:45 | 15.00 | unsuccessful | Q004 | C003 | D003 |
+-----+-----+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

–Deleting a canceled order

DELETE FROM Order

WHERE OrderID = '0003';

–View the order table after updated

SELECT * FROM Order;

```
mysql> SELECT * FROM `Order`;
+-----+-----+-----+-----+-----+-----+-----+-----+
| OrderID | FruitType | OrderDate | TotalAmount | PaymentStatus | QueueNumber | CustomerID | DriverID |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 0001 | apple | 2024-12-22 00:01:53 | 12.00 | successful | Q001 | C001 | D001 |
| 0002 | melon | 2024-12-22 00:05:00 | 20.00 | successful | Q002 | C002 | D002 |
| 0004 | pears | 2024-12-22 00:07:45 | 15.00 | unsuccessful | Q004 | C003 | D003 |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

–Make payment

INSERT INTO Payment (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)

VALUES ('P001', 'OnlinePayment', 'C001', 'SE001', 'S001');

INSERT INTO Payment (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)

VALUES ('P002', 'OnlinePayment', 'C002', 'SE001', 'S001');

```
INSERT INTO Payment (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)
VALUES ('P003', 'Cash', 'C001', 'SE002', 'S002');
```

```
INSERT INTO Payment (PaymentID, PaymentMethod, CustomerID, SellerID, SupplierID)
VALUES ('P004', 'OnlinePayment', 'C003', 'SE003', 'S003');
```

–View Payment table

```
SELECT * FROM Payment;
```

```
mysql> SELECT * FROM Payment;
+-----+-----+-----+-----+-----+
| PaymentID | PaymentMethod | CustomerID | SellerID | SupplierID |
+-----+-----+-----+-----+-----+
| P001      | OnlinePayment | C001      | SE001    | S001      |
| P002      | OnlinePayment | C002      | SE001    | S001      |
| P003      | Cash         | C001      | SE002    | S002      |
| P004      | OnlinePayment | C003      | SE003    | S003      |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

–Delete wrong payment

```
DELETE FROM Payment
```

```
WHERE PaymentID = 'P004';
```

–View Payment table

```
SELECT * FROM Payment;
```

```
mysql> SELECT * FROM Payment;
```

PaymentID	PaymentMethod	CustomerID	SellerID	SupplierID
P001	OnlinePayment	C001	SE001	S001
P002	OnlinePayment	C002	SE001	S001
P003	Cash	C001	SE002	S002

```
3 rows in set (0.00 sec)
```

–Online Payment

```
INSERT INTO OnlinePayment (OnlinePaymentID, AmountPaid, PaymentDate)
```

```
VALUES ('ON001', 12.00, SYSDATE());
```

```
INSERT INTO OnlinePayment (OnlinePaymentID, AmountPaid, PaymentDate)
```

```
VALUES ('ON002', 20.00, SYSDATE());
```

–View Online payment table

```
SELECT * FROM Online payment;
```

```
mysql> SELECT * FROM OnlinePayment;
```

OnlinePaymentID	AmountPaid	PaymentDate
ON001	12	2024-12-21 22:10:08
ON002	20	2024-12-21 22:12:48

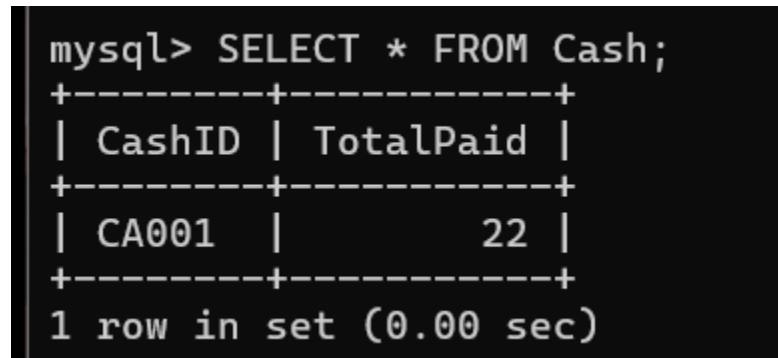
```
2 rows in set (0.00 sec)
```

–Cash

```
INSERT INTO Cash (CashID, TotalPaid)
VALUES ('CA001', 22);
```

–View Cash table

```
SELECT * FROM Cash;
```



```
mysql> SELECT * FROM Cash;
+-----+-----+
| CashID | TotalPaid |
+-----+-----+
| CA001  |         22 |
+-----+-----+
1 row in set (0.00 sec)
```

–Change online payment to cash

```
DELETE FROM Online Payment
```

```
WHERE OnlinePaymentID = 'ON002';
```

```
INSERT INTO Cash (CashID, TotalPaid)
```

```
VALUES ('CA002', 22);
```

–View the Cash and Online payment table

```
SELECT * FROM Cash;
```

```
SELECT * FROM Online payment;
```

```
mysql> SELECT * FROM Cash;
+-----+-----+
| CashID | TotalPaid |
+-----+-----+
| CA001  |          22 |
| CA002  |          22 |
+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM OnlinePayment;
+-----+-----+-----+
| OnlinePaymentID | AmountPaid | PaymentDate |
+-----+-----+-----+
| ON001           |          12 | 2024-12-21 22:10:08 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

–Inventory

```
INSERT INTO Inventory (InventoryID, FruitType, QuantityAvailable, SellerID)
VALUES ('IR001', 'apple', 100, 'SE001');
```

```
INSERT INTO Inventory (InventoryID, FruitType, QuantityAvailable, SellerID)
VALUES ('IR002', 'mango', 100, 'SE002');
```

```
INSERT INTO Inventory (InventoryID, FruitType, QuantityAvailable, SellerID)
VALUES ('IR003', 'banana', 100, 'SE002');
```

–View Inventory table

```
SELECT * FROM Inventory;
```

```
mysql> SELECT * FROM Inventory  
-> ;
```

InventoryID	FruitType	QuantityAvailable	SellerID
IR001	apple	100	SE001
IR002	mango	100	SE002
IR003	banana	100	SE002

3 rows in set (0.00 sec)

-Update Inventory

UPDATE Inventory

SET FruitType = 'Orange', QuantityAvailable = 110, SellerID = 'SE002'

WHERE InventoryID = 'IR002';

-View Inventory table

SELECT * FROM Inventory;

```
mysql> SELECT * FROM Inventory;
```

InventoryID	FruitType	QuantityAvailable	SellerID
IR001	apple	100	SE001
IR002	orange	110	SE002
IR003	banana	100	SE002

3 rows in set (0.00 sec)

-Invoice

INSERT INTO Invoice(InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)

VALUES ('I001', SYSDATE(), 200, 'O001', 'S001');

```
INSERT INTO Invoice (InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)
VALUES ('I002', SYSDATE(), 150, 'O001', 'S002');
```

```
INSERT INTO Invoice(InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)
VALUES ('I003', SYSDATE(), 200, 'O004', 'S001');
```

```
INSERT INTO Invoice(InvoiceID, InvoiceDate, TotalAmount, OrderID, SellerID)
VALUES ('I004', SYSDATE(), 250, 'O004', 'S003');
```

–View Invoice table

```
SELECT * FROM Invoice;
```

```
mysql> SELECT * FROM Invoice;
+-----+-----+-----+-----+-----+
| InvoiceID | InvoiceDate       | TotalAmount | OrderID | SellerID |
+-----+-----+-----+-----+-----+
| I001     | 2024-12-22 00:16:08 | 200.00     | 0001    | SE001    |
| I002     | 2024-12-22 00:17:19 | 150.00     | 0001    | SE002    |
| I003     | 2024-12-22 00:17:19 | 200.00     | 0002    | SE001    |
| I004     | 2024-12-22 00:17:19 | 250.00     | 0002    | SE003    |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

–Update Invoice

```
UPDATE Invoice
```

```
SET TotalAmount = 170
```

```
WHERE InvoiceID = 'I003';
```

–View Invoice table

SELECT * FROM Invoice;

```
mysql> SELECT * FROM Invoice;
```

InvoiceID	InvoiceDate	TotalAmount	OrderID	SellerID
I001	2024-12-22 00:16:08	200.00	0001	SE001
I002	2024-12-22 00:17:19	150.00	0001	SE002
I003	2024-12-22 00:17:19	170.00	0002	SE001
I004	2024-12-22 00:17:19	250.00	0002	SE003

4 rows in set (0.00 sec)

-Feedback

INSERT INTO Feedback (FeedbackID, Message, Date, CustomerID)

VALUES

('F001', 'Great service and fresh fruits!', '2024-12-20 10:00:00', 'C001'),

('F002', 'Delivery was fast and smooth.', '2024-12-20 11:00:00', 'C002'),

('F003', 'Amazing quality and reasonable prices.', '2024-12-20 12:00:00', 'C003');

–View Feedback table

SELECT * FROM Feedback;

FeedbackID	Message	Date	CustomerID
F001	Great service and fresh fruits!	2024-12-20 10:00:00	C001
F002	Delivery was fast and smooth.	2024-12-20 11:00:00	C002
F003	Amazing quality and reasonable prices.	2024-12-20 12:00:00	C003

7.0 Summary

In this project phase, we converted the conceptual Entity Relationship Diagram (ERD) to a Logical ERD to ensure full compliance with relational database principles by handling non-relational features and refining complex relationships. The logical ERD reflects a structured and efficient representation of system entities with explicit attributes, primary keys, and relationships customized according to business rules.

Next, we created a relational schema based on the logical ERDs and normalized it to BCNF to avoid duplicate data and ensure the database was well organized. We also updated the data dictionary to match the new structure, keeping all definitions accurate and consistent. The final logical ERD shows a better database design, ensuring that data can be stored and managed efficiently. We also wrote SQL statements (DDL and DML) to create and manage the database tables to ensure the design supported the system requirements.

In conclusion, this phase focuses on creating a simple and user-friendly system to help traditional business owners, especially those unfamiliar with technology. This system will simplify inventory management by showing real-time stock updates and tracking sales and profits. It helps solve problems such as marketing and limited resources, allowing business owners to improve their work and compete in the digital age.

Interface

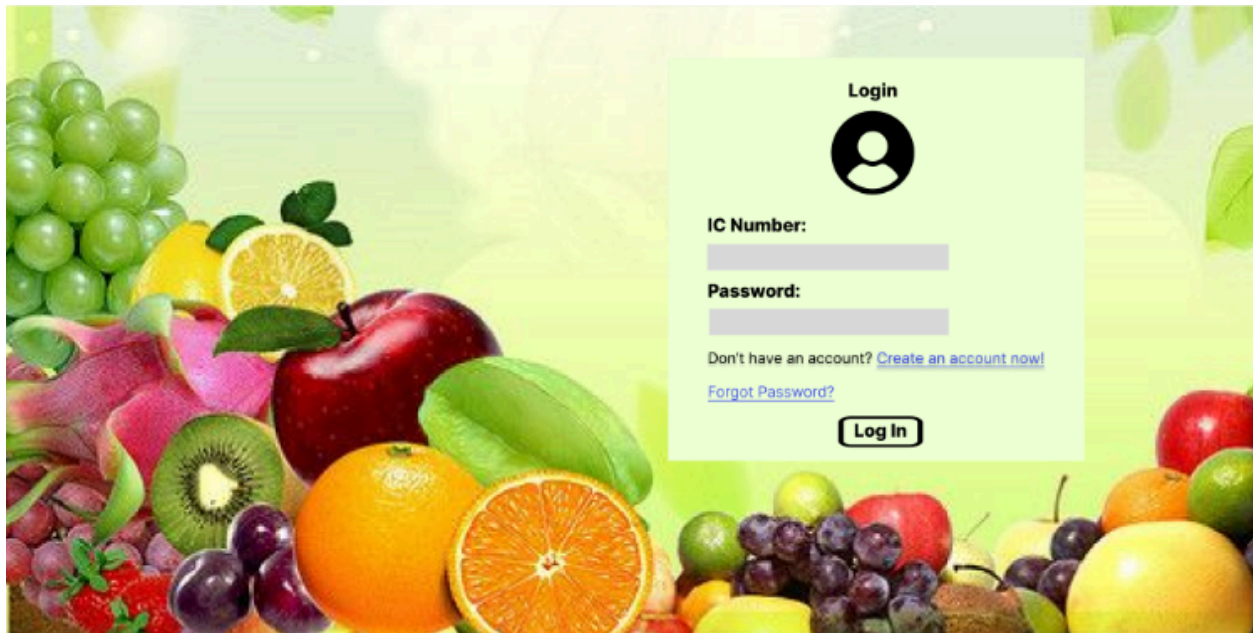


Figure 1.1 Login or register system page

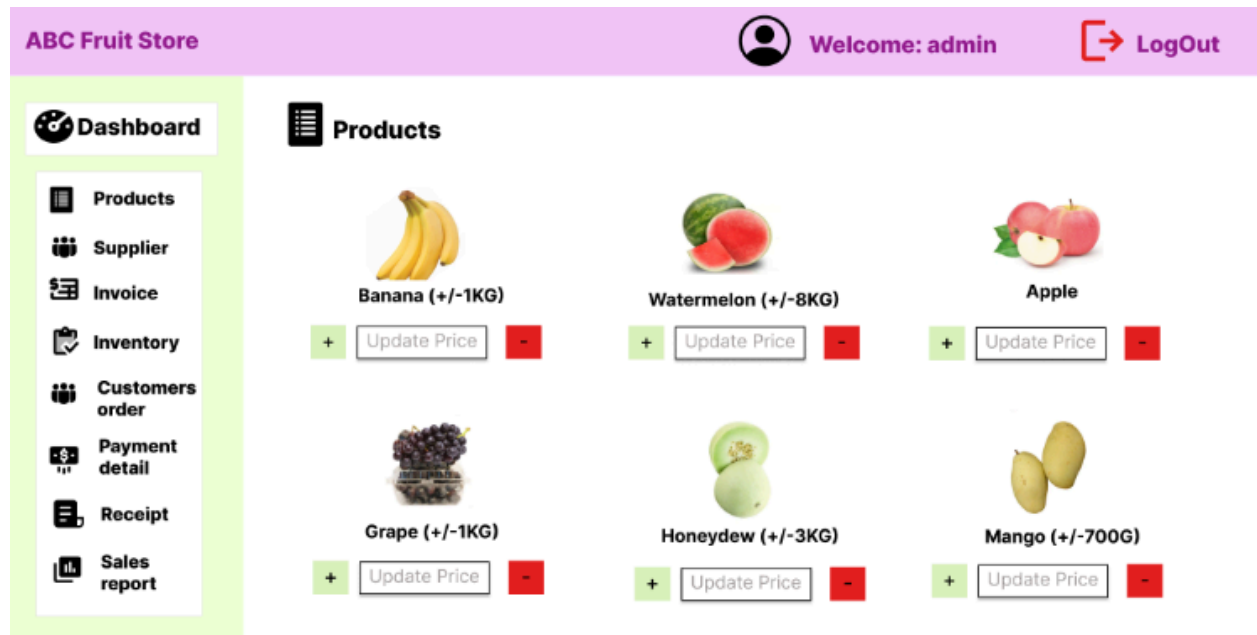


Figure 1.2 Update the price for each of the products

ABC Fruit Store

Welcome: admin

LogOut

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Products

Categories

☐ Banana
☐ Watermelon
☐ Apples
☐ Grapes
☐ Honeydews
☐ Mango

Banana (+/-1KG)
RM7.20

Watermelon (+/-8KG)
RM40.00

Apple
RM 3.50

Grape (+/-1KG)
RM20.20

Honeydew (+/-3KG)
RM15.00

Mango (+/-700G)
RM 9.30

Figure 1.3 Products list and price that ABC Fruit Store sold

ABC Fruit Store

Welcome: admin

LogOut

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Supplier

Order	Date oder	Order Quantity	Received Quantity	Pending Quantity	Overall Status
Banana	11/6/2024	130	130	0	COMPLETED
Watermelon	11/6/2024	200	0	200	PENDING
Apple	11/6/2024	80	55	25	PENDING
Grape	11/6/2024	60	60	0	COMPLETED
Honeydew	11/6/2024	0	0	0	COMPLETED
Mango	11/6/2024	50	30	20	PENDING

Figure 1.4 Purchase Order from Supplier

ABC Fruit Store

Welcome: admin

Logout

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Invoice

Month

☐ January
☐ February
☐ March
☐ April
☐ May
☒ June
☐ July
☐ August
☐ September
☐ October
☐ November
☐ December

Invoice June

Billed To:

Mr Lim
M16 Ktdi,
Skudai UTM,
81310 Johor Bahru,
Johor

Invoice #	37482
Date	13/6/2024
Total	RM 800

Product	Quantity	Price (RM)	Amount (RM)
1. Banana	20	7.20	144
2. Watermelon	30	40.00	1200
3. Apples	50	3.50	175
4. Grapes	20	20.20	404
5. Honeydews	40	15.00	600
6. Mango	40	9.30	279
			Total: 2802

Figure 1.5 Invoice provide by supplier

ABC Fruit Store

Welcome: admin

Logout

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Inventory

Update product

Stock in

Stock out

Update status

Submit

Figure 1.6 Update the products stock and status

ABC Fruit Store

Welcome: admin

LogOut

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Inventory

Product	Date	Status	Stock in	Stock out	Stock balance
Banana	12/6/2024	In Stock	130	45	85
Watermelon	12/6/2024	Out of Stock	0	0	0
Apple	12/6/2024	In Stock	80	55	25
Grape	12/6/2024	In Stock	60	35	25
Honeydew	12/6/2024	Out of Stock	0	0	0
Mango	12/6/2024	In Stock	50	20	30

Figure 1.7 Inventory of ABC Fruit Store

ABC Fruit Store

Welcome: admin

LogOut

Dashboard

Products

Supplier

Invoice

Inventory

Customers order

Payment detail

Receipt

Sales report

Customers order

Customers' Name

Contact

Date order

Order

1. Alice Lee

0129402403

12/6/2024

Banana -2 , Apple -3

3. Kang Huey

0179402947

12/6/2024

Honeydew - 3

4. Sze Qing

0123067293

12/6/2024

Grape -1 , Apple -2 , Mango -2

5. Zi Ling

0110572830

12/6/2024

Mango -3 , Grape -2

6. Zhi Yu

0133058172

12/6/2024

Honeydew -1 , Watermelon -1

Figure 1.8 Customer order detail

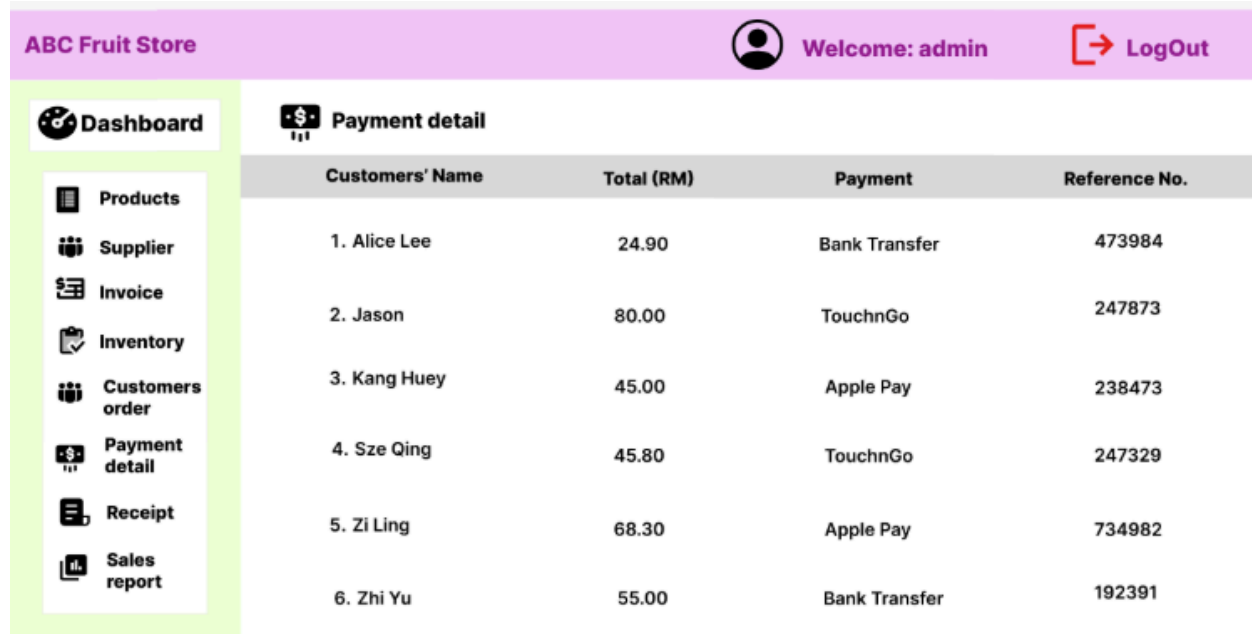


Figure 1.9 Payment detail done by customers

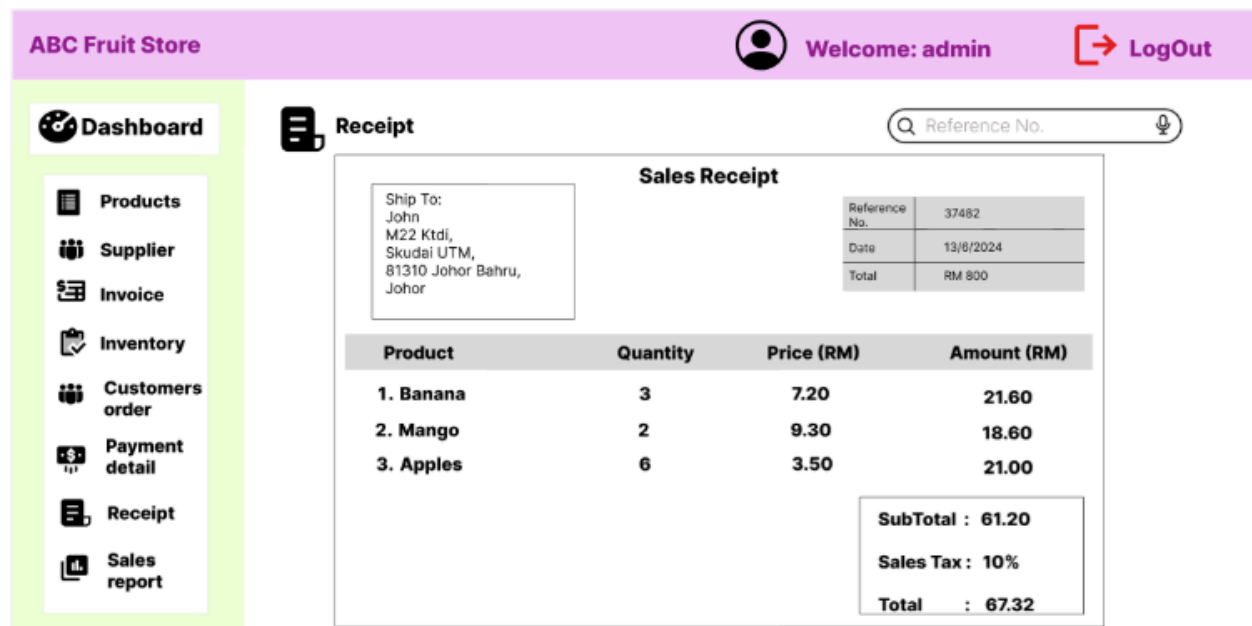


Figure 1.10 Receipt given to customers

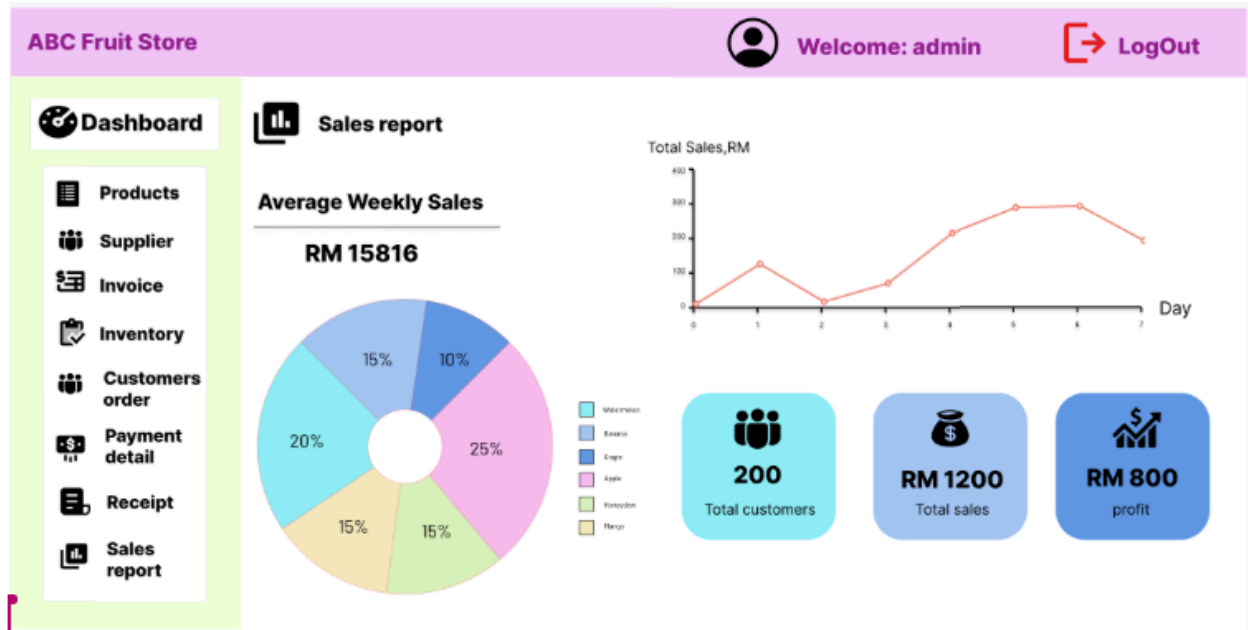


Figure 1.11 Average Weekly Sales Report for ABC Fruit Store